Arithmetic Certificates and Computational Evidence for the CM Packet and Quotient Papers

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3 November 2026

Abstract

This supplement records the arithmetic domain, orientation and module calculations, retained geometric exclusion batches, auxiliary lattice constructions and execution provenance for two papers on a fixed CM packet and a finite quotient family. It separates mathematical exhaustiveness, exact decision procedures and historical execution assumptions. A subsidiary comparison explains the normalization and Hecke-search issues relevant to prior work. The document provides an guide to the retained research evidence; it does not claim a new full-suite replay or a complete public evidence archive.

Companion references. Result numbers prefixed I, II, H and S refer respectively to the packet paper, quotient paper, supporting Hodge note and computational supplement. Each document has its own numbering and bibliography.

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1 How to use this supplement

This supplement contains the detailed packet arithmetic, historical exclusion records, auxiliary lattice arguments and execution provenance used by Papers I and II and the supporting Hodge note. It is organized by mathematical role. The historical partial computations are explicitly separated from the final complete packet theorem in Paper I.

Notation in packet sections is $M = \mathbb{Q}(\zeta_{42})$, $L = M^+$, $t = \zeta_{42} + \zeta_{42}^{-1}$ and $\pi = t^4 + t^3 - 4t^2 - 2t + 5$ of norm 43. In quotient-family material, π instead denotes $1 - z^{14}$, the generator over 3 specified in Paper II; these symbols must not be identified. Embeddings are ordered $1, 5, 11, 13, 17, 19$; $K = \mathbb{Q}(\sqrt{-3})$.

The roles of evidence are distinct: a finite-domain proof establishes exhaustiveness; exact arithmetic or outward enclosures justify decisions; source/runtime and retained-output records describe the execution to which the claim is bound. A hash establishes file identity, not mathematical correctness or absence of a historical transient replacement.

1.1 Result-to-evidence guide

Packet domain. Paper I, Theorem I.3.3, uses the order and module calculations below and the full-packet arithmetic sources in Section 13.

Curve intersection. Paper I, Theorem I.3.6, uses its two-region lemma, positive vertex realizations and both complete negative populations. The final receipt is `experiments/continuous_v83/whole_packet_assessment/final-adoption.json`. The historical execution premise in Paper I remains required.

Principal geometry. The common-lattice, stabilizer and map proofs are in Paper I. Their named exact regressions and the supplementary individual-lattice arguments below do not replace those proofs.

Finite quotients and descent. Paper II names the adopted exact finite tables within its proofs. Subgroup, underlying, polarized, orbit and field counts are different assertions, with different proof inputs.

Jacobian exclusion. Paper II, Theorem II.12.4, retains its model, theta-enclosure and execution contract. Counts cannot be pooled across the four models. The 156 smaller-order cases remain open.

Hodge tensors. The Hodge note gives the algebraic constructions and integral relation arguments; source identity alone does not prove either.

This is a navigation guide, not a newly completed transitive evidence archive. The repository remains private. Historical source paths are relative to its root; a release must provide the complete dependencies and accurate access instructions before claiming reproducibility.

2 Detailed arithmetic underlying the $k = 11$ certificate

This supplement contains the derivational arithmetic suppressed from the main proof path. It also records the broader 42-trace computation from which the $k = 11$ point was isolated.

3 The prime 43: an explicit totally positive generator

We now specialize to the first completely split rational prime, 43. Consider

$$\mathfrak{l} = (43, t - 5) \subset \mathcal{O}_L.$$

Define

$$\alpha_0 = 1 + \zeta_{42} + \zeta_{42}^4 \in \mathcal{O}_M. \tag{3.1}$$

A direct resultant computation gives

$$N_{M/\mathbb{Q}}(\alpha_0) = 43. \tag{3.2}$$

Its relative norm is

$$\begin{aligned} \pi &= \alpha_0 \bar{\alpha}_0 && (3.3)\\ &= 3 + (\zeta + \zeta^{-1}) + (\zeta^3 + \zeta^{-3}) + (\zeta^4 + \zeta^{-4}) && (3.4)\\ &= t^4 + t^3 - 4t^2 - 2t + 5. && (3.5) \end{aligned}$$

The resultant of $f(T)$ in (I.2.1) with the polynomial in (3.5) is 43, hence

$$N_{L/\mathbb{Q}}(\pi) = 43.$$

Moreover π is totally positive because for every real embedding σ and either extension $\widetilde{\sigma} : M \hookrightarrow \mathbb{C}$,

$$\sigma(\pi) = |\widetilde{\sigma}(\alpha_0)|^2 > 0.$$

Finally $\pi(5) \equiv 0 \pmod{43}$, so $(\pi) = \mathfrak{l}$.

Proposition 3.1. *The element*

$$\pi_1 = t^4 - t^2 + t - 3$$

listed in [6, Sec. 5.3] has norm 43 *but is not totally positive.*

Proof. At the embeddings indexed by $k = 11$ and 13 one obtains, respectively,

$$\sigma_{11}(\pi_1) \approx -3.1712995, \qquad \sigma_{13}(\pi_1) \approx -3.9795331.$$

The signs can be certified with rational isolating intervals for the roots of (I.2.1). □

Remark 3.2. If $M_2^+(\mathcal{O}_L)$ is intended in its standard Hilbert-modular sense of matrices with totally positive determinant, then the representatives $\mathrm{diag}(\pi_1, 1)$ cannot lie in $M_2^+(\mathcal{O}_L)$ as written. The totally positive generator (3.5) repairs this issue for the chosen prime.

Remark 3.3 (A related matrix-group typo). Section 5.1 says that $\sigma_i(\gamma)$ lies in $\mathrm{SL}_2(\mathbb{R})$ while simultaneously imposing $\det(\gamma)\mathcal{O}_L = \mathfrak{l}$. For a genuine $\mathfrak{l}$-Hecke matrix one has $\det(\sigma_i(\gamma)) = \sigma_i(\pi)$, not 1. The Möbius action is of course well-defined for $\mathrm{GL}_2^+(\mathbb{R})$ (or projectively), so this is repairable, but it reinforces that the determinant is the local element π.

4 Classification of the integral norm fiber

The cyclotomic structure makes the corrected norm equation completely explicit.

Theorem 4.1 (The 84 elements and 42 traces)**.** *Let π and α_0 be as in (3.5) and (3.1). Then*

$$\{x \in \mathcal{O}_M : x\bar{x} = \pi\} = \{\zeta_{42}^k \alpha_0 : 0 \le k < 42\} \sqcup \{\zeta_{42}^k \bar{\alpha}_0 : 0 \le k < 42\}. \tag{4.1}$$

In particular the norm fiber contains exactly 84 elements. Modulo complex conjugation it has 42 classes, and the corresponding relative traces

$$c_k = \zeta_{42}^k \alpha_0 + \zeta_{42}^{-k} \bar{\alpha}_0 \in \mathcal{O}_L$$

are pairwise distinct.

Proof. Because $(\alpha_0) = \mathfrak{P}$ is a prime of M above $\mathfrak{l}$ and

$$(\pi) = \mathfrak{P}\bar{\mathfrak{P}},$$

any integral x with $(x)(\bar{x}) = (\pi)$ must have $(x) = \mathfrak{P}$ or $(x) = \bar{\mathfrak{P}}$. Thus $x = u\alpha_0$ or $x = u\bar{\alpha}_0$ for a unit u satisfying $u\bar{u} = 1$.

If $u\bar{u} = 1$, then for every complex embedding $\tau : M \hookrightarrow \mathbb{C}$ one has $|\tau(u)|^2 = \tau(u\bar{u}) = 1$. Since u is an algebraic integer, Kronecker's theorem implies that u is a root of unity. Conversely every root of unity has relative norm one. The root-of-unity group of $M = \mathbb{Q}(\zeta_{42})$ is μ_{42}, so the norm-one unit kernel is exactly μ_{42}, giving (4.1). The two displayed sets are disjoint because they generate the distinct prime ideals $\mathfrak{P}$ and $\bar{\mathfrak{P}}$.

Conjugation pairs the 84 elements. If two conjugacy classes had the same trace, their representatives would have the same trace and the same norm π, hence would be roots of the same quadratic $X^2 - cX + \pi$; they would therefore be equal or conjugate. Thus there are exactly 42 distinct traces. □

For example,

$$c_0 = \alpha_0 + \bar{\alpha}_0 = t^4 - 4t^2 + t + 4.$$

This theorem replaces an unstructured search for the relative-norm stage by a finite and complete classification for the chosen split prime.

5 Certified signature enumeration

For $d = 3$ and $d = 7$, the same cyclotomic field can be written as $M = L(\sqrt{-d})$. We choose

$$\sqrt{-3} = 1 + 2\zeta_{42}^{14}$$

and

$$\sqrt{-7} = 1 + 2(\zeta_{42}^{6} + \zeta_{42}^{12} + \zeta_{42}^{24}),$$

which square to -3 and -7, respectively.

For

$$\alpha_k = \zeta_{42}^k \alpha_0,$$

define $a_{k,d}, b_{k,d} \in L$ by

$$\alpha_k = a_{k,d} + b_{k,d}\sqrt{-d}.$$

The signs of the six real embeddings of $b_{k,d}$ can be certified exactly by expressing $b_{k,d}$ as a degree-at-most-five polynomial in t, isolating the six real roots of (I.2.1) with rational intervals, and evaluating by rational interval arithmetic.

Theorem 5.1 (Arithmetic $(3,3)$ signature counts)**.** *With the normalization above:*

(a) for $d = 3$, exactly 24 of the 42 trace classes have $b_{k,3}$ with three positive and three negative real embeddings;

(b) for $d = 7$, exactly 4 of the 42 trace classes have $b_{k,7}$ with three positive and three negative real embeddings.

A set of representatives is

$$d = 3: \quad k \in \{0, 1, 4, 8, 11, 12, 15, 16, 17, 18, 19, 20, 21, 22, 25, 29, \\ 32, 33, 36, 37, 38, 39, 40, 41\},$$

and

$$d = 7: \quad k \in \{2, 14, 23, 35\}.$$

One explicit $d = 3$ example is

$$b_{0,3} = -\frac{1}{3}t^5 - \frac{1}{6}t^4 + \frac{7}{3}t^3 + t^2 - \frac{23}{6}t - \frac{4}{3},$$

whose signs at $k = 1, 5, 11, 13, 17, 19$ are

$$(+, -, -, +, +, -).$$

For $d = 7$, one example is

$$b_{2,7} = -\frac{5}{14}t^5 + \frac{31}{14}t^3 + \frac{3}{7}t^2 - \frac{45}{14}t - \frac{15}{14},$$

with sign vector

$$(+, -, -, +, -, +).$$

Remark 5.2 (What Theorem 5.1 does not prove). The sign vector of b is an arithmetic compatibility condition, not the CM type of a period point. The preprint itself notes that the Hecke endomorphism alone does not determine the CM type. Therefore Theorem 5.1 does not prove

$$V \cap \mathcal{W}_K \neq \varnothing.$$

It proves only that the corrected relative-norm fiber contains integral CM elements whose b-coefficients have the desired $3 + 3$ sign pattern.

6 Canonical fixed points and a finite replacement for the lift variable

The missing elimination of δ in Section 5.3 can be reorganized using integral conjugacy rather than by ranging directly over $\mathrm{SL}_2(\mathcal{O}_L)$. For any corrected trace $c \in \mathcal{O}_L$ define the companion matrix

$$A_c = \begin{pmatrix} 0 & -\pi \\ 1 & c \end{pmatrix} \in M_2(\mathcal{O}_L). \tag{6.1}$$

Then $\det A_c = \pi$ and $\chi_{A_c}(X) = X^2 - cX + \pi$.

Proposition 6.1 (Canonical Hecke-fixed CM representative)**.** *Assume $c^2 - 4\pi$ is totally negative. For each real embedding $\sigma_i : L \hookrightarrow \mathbb{R}$, put*

$$w_i(c) = \frac{-\sigma_i(c) + \sqrt{\sigma_i(c)^2 - 4\sigma_i(\pi)}}{2}, \tag{6.2}$$

where the square root is chosen with positive imaginary part. Then

$$w(c) = (w_1(c), \dots, w_6(c)) \in \mathbb{H}^6$$

is fixed componentwise by A_c under the fractional-linear action. Thus A_c supplies an explicit determinant-π Hecke-fixed CM representative in the Hilbert modular domain.

Proof. The fixed-point equation for $\left(\begin{smallmatrix} 0 & -\pi \\ 1 & c \end{smallmatrix}\right)$ is $w = -\pi/(w + c)$, equivalently $w^2 + cw + \pi = 0$. Applying σ_i gives (6.2). Total negativity of the discriminant puts the selected root in $\mathbb{H}$ at every embedding. The characteristic polynomial generates the same quadratic CM field as a root of $X^2 - cX + \pi$. □

For the 42 traces of Theorem 4.1, Proposition 6.1 therefore gives 42 explicit canonical fixed-point representatives. They need not be pairwise distinct in the quotient X_L, and they do not automatically exhaust all integral conjugacy classes with the same characteristic polynomial.

That last caveat is exactly where the infinite δ-variable belongs. Generalized Latimer–MacDuffee theory replaces conjugacy of integral matrices with a finite module-classification problem for the quadratic order

$$R_c = \mathcal{O}_L[\alpha], \qquad \alpha^2 - c\alpha + \pi = 0.$$

Generalized Latimer–MacDuffee theory identifies the analogous integral problem with module classes over the quadratic order. For comparison, algorithms for prescribed characteristic and minimal polynomials over $\mathbb{Z}$ are developed in [1]. In the maximal-order cases needed here, the $\mathcal{O}_L$-module classification is handled directly below using class number one. Consequently, a corrected finite search should proceed trace by trace through the finitely many integral conjugacy classes attached to R_c, rather than count root choices before the lift has been eliminated.

McMullen's standard modular embedding has the normalization

$$F(z) = (z, f_2(z), \dots, f_6(z)) \tag{6.3}$$

with equivariance $f_i(g_1 z) = g_i f_i(z)$ [3, Sec. 6]. For the canonical representative (6.1), the concrete identities

$$f_i(w_1(c)) = w_i(c), \qquad i = 2, \dots, 6 \tag{6.4}$$

are therefore a sufficient test that the chosen lift $w(c)$ itself lies on the distinguished lift $F(\mathbb{H}) \subset \mathbb{H}^6$.

There is an important quotient-level qualification. If Γ_L denotes the Hilbert modular group defining X_L, then the full inverse image of the McMullen curve in $\mathbb{H}^6$ is the Γ_L-saturation

$$\bigcup_{\gamma \in \Gamma_L} \gamma\, F(\mathbb{H}).$$

Consequently the complete membership criterion for the quotient point $[w(c)] \in X_L$ is

$$\exists\, \gamma \in \Gamma_L,\ z \in \mathbb{H} \quad \text{such that} \quad \gamma \cdot w(c) = F(z). \tag{6.5}$$

Equation (6.4) is the special case $\gamma = 1$. For a non-companion integral conjugacy representative one obtains the analogous distinguished-lift test after the corresponding $\mathrm{SL}_2(\mathcal{O}_L)$ normalization, but a complete quotient-level search must still account for the orbit condition (6.5). Thus the arithmetic reductions below make the set of Hecke conjugacy classes finite; they do not by themselves collapse the geometric orbit problem to the single lift $F(\mathbb{H})$.

7 Component normalization and an effective finite orbit search

There is a further normalization issue that must be separated from the arithmetic corrections above. McMullen's construction first produces the modular embedding in the disconnected Hilbert modular variety

$$Y_L = (\pm\mathbb{H})^6/\mathrm{SL}_2(\mathcal{O}_L),$$

not directly in $X_L = \mathbb{H}^6/\mathrm{SL}_2(\mathcal{O}_L)$. If a Galois-conjugate triangle has opposite orientation, its component map takes values in $-\mathbb{H}$ while still satisfying the equivariance relation (12.6); McMullen then uses commensurability of the components of Y_L and passes to a finite-index subgroup to obtain a curve on X_L [3, Prop. 6.1, Cor. 6.2 and proof of Thm. 1.9].

The shortcut in [6, Sec. 1.4] of replacing a map $f_i : \mathbb{H} \to -\mathbb{H}$ by $-\overline{f_i}$ does not preserve the same target representation in general. Let

$$J(z) = -\overline{z}, \qquad s = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \in \mathrm{PGL}_2(\mathbb{R}).$$

For $g \in \mathrm{PSL}_2(\mathbb{R})$ one has

$$J \circ g \circ J^{-1} = sgs^{-1},$$

which is not generally equal to g. Hence if $f_i(g_1 z) = g_i f_i(z)$, then $J \circ f_i$ is equivariant for $g_i \mapsto s g_i s^{-1}$, not for the original g_i. For example, with $B = \left(\begin{smallmatrix} 0 & -1 \\ 1 & t \end{smallmatrix}\right)$,

$$J(Bz) = \frac{1}{\overline{z} + t}, \qquad B(Jz) = \frac{1}{\overline{z} - t},$$

so the two actions differ for $t \neq 0$.

Consequently an exact Hecke search on the curve in X_L must specify the finite-cover and component-changing normalization used to pass from McMullen's Y_L-curve to X_L. This does not alter the abstract local norm calculations of the preceding sections, but it is part of the data required to identify an arithmetic Hecke class with a point on the actual curve in X_L.

The missing normalization can be written explicitly. Let $\varepsilon_i \in \{\pm 1\}$ record the component containing $f_i(\mathbb{H})$, so that $f_i(\mathbb{H}) \subset \varepsilon_i \mathbb{H}$ and $\varepsilon_1 = 1$. By weak approximation choose $\lambda \in L^\times$ satisfying $\mathrm{sgn}(\sigma_i(\lambda)) = \varepsilon_i$ for all i, and put

$$h = \begin{pmatrix} \lambda & 0 \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2(L).$$

Then the i-th real component of h sends $\varepsilon_i \mathbb{H}$ to $\mathbb{H}$. Define

$$\Gamma_\lambda = \Delta_0 \cap h^{-1}\mathrm{SL}_2(\mathcal{O}_L)h. \tag{7.1}$$

Since $h\mathrm{SL}_2(\mathcal{O}_L)h^{-1}$ is commensurable with $\mathrm{SL}_2(\mathcal{O}_L)$, the subgroup Γ_λ has finite index in Δ_0. For

$$F_\lambda(z) = h \cdot F(z) \in \mathbb{H}^6$$

and $g \in \Gamma_\lambda$, equivariance becomes

$$F_\lambda(g_1 z) = (hgh^{-1}) \cdot F_\lambda(z), \qquad hgh^{-1} \in \mathrm{SL}_2(\mathcal{O}_L).$$

Thus F_λ descends to the finite-cover curve $\mathbb{H}/\Gamma_\lambda \to X_L$. This is the concrete specialization of McMullen's commensurability argument in Proposition 6.1 and Corollary 6.2. In particular, after λ is fixed the Hecke-orbit test must be performed against F_λ, not against an ad hoc componentwise conjugation of F.

For the present $(14, 21, 42)$ example the component correction is especially simple and does not itself force a smaller subgroup.

Proposition 7.1 (Explicit component orientation and integral transfer)**.** *Order the six embeddings by $k = 1, 5, 11, 13, 17, 19$. The orientations of the six Galois triangles relative to the identity triangle are*

$$(\varepsilon_1, \dots, \varepsilon_6) = (+, +, -, -, -, -).$$

The unit

$$t = 2\cos(\pi/21) \in \mathcal{O}_L^\times$$

has the same real sign vector. Consequently, with

$$h_t = \begin{pmatrix} t & 0 \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2(\mathcal{O}_L),$$

the map $F_t = h_t \cdot F$ takes values in $\mathbb{H}^6$, and $h_t \mathrm{SL}_2(\mathcal{O}_L) h_t^{-1} = \mathrm{SL}_2(\mathcal{O}_L)$. Thus the component-changing normalization may be carried out integrally on X_L itself; no additional finite-index subgroup is required solely for the purpose of changing the signs of the six half-plane components.

Proof. Use McMullen's printed matrices $A, B, C \in \mathrm{SL}_2(\mathcal{O}_L)$ for three consecutively ordered vertices of the quadrilateral model of Δ_0. For each of the six real embeddings indexed by $k = 1, 5, 11, 13, 17, 19$, let $p_A, p_B, p_C \in \mathbb{H}$ be their upper-half-plane fixed points. The oriented Euclidean determinant

$$\Omega_k = \det \begin{pmatrix} \Re(p_B - p_A) & \Im(p_B - p_A) \\ \Re(p_C - p_A) & \Im(p_C - p_A) \end{pmatrix}$$

has certified rational-interval signs

$$\Omega_1 < 0, \quad \Omega_5 < 0, \quad \Omega_{11} > 0, \quad \Omega_{13} > 0, \quad \Omega_{17} > 0, \quad \Omega_{19} > 0.$$

The six isolating intervals are disjoint and each has an exact sign change for the degree-six minimal polynomial, so they contain all six real roots, one apiece. Interval evaluation of Ω_k then excludes zero in every case. Thus, relative to $k = 1$, the orientation vector is $(+, +, -, -, -, -)$. The same sign vector for t is immediate from $\sigma_k(t) = 2\cos(k\pi/21)$. Finally, the minimal polynomial (I.2.1) has constant term 1, so $N_{L/\mathbb{Q}}(t) = 1$ and $t \in \mathcal{O}_L^\times$. Hence $h_t \in \mathrm{GL}_2(\mathcal{O}_L)$ and conjugation by h_t preserves $\mathrm{SL}_2(\mathcal{O}_L)$. □

A torsion-free or level cover may still be imposed for fine-moduli purposes, but the component transfer itself is already integral. This removes one previously unspecified normalization from the Hecke comparison: in what follows the normalized lift can be taken to be F_t.

Once that normalization is fixed, compactness gives an effective replacement for the apparently infinite orbit search in (6.5).

Proposition 7.2 (Effective finiteness of the orbit test)**.** *Let $G \subset \mathrm{SL}_2(L)$ be an arithmetic subgroup commensurable with $\mathrm{SL}_2(\mathcal{O}_L)$ acting diagonally on $\mathbb{H}^6$, let $F : \mathbb{H} \to \mathbb{H}^6$ be a Γ-equivariant modular embedding with Γ cocompact, and let $D \subset \mathbb{H}$ be a compact fundamental polygon for Γ. Put $K = F(D)$. For every fixed $w = (w_1, \dots, w_6) \in \mathbb{H}^6$, the set*

$$\{\gamma \in G : \gamma w \in K\}$$

is finite. Moreover, after choosing an integral lattice containing G, it is effectively enumerable from certified lower bounds for the coordinate imaginary parts on K and upper bounds for the coordinate absolute values on K.

Proof. Write $w_i = x_i + iy_i$ with $y_i > 0$, and $\gamma_i = \left(\begin{smallmatrix} a_i & b_i \\ c_i & d_i \end{smallmatrix}\right)$ for the i-th real embedding of γ. Since K is compact in $\mathbb{H}^6$, there are constants

$$m_i = \inf_{z \in K} \operatorname{Im}(z_i) > 0, \qquad U_i = \sup_{z \in K} |z_i| < \infty.$$

If $u = \gamma w \in K$, the Möbius imaginary-part identity gives

$$\operatorname{Im}(u_i) = \frac{y_i}{|c_i w_i + d_i|^2} \geq m_i,$$

hence

$$|c_i w_i + d_i| \leq B_i := \sqrt{y_i / m_i}.$$

Taking imaginary parts yields

$$|c_i| \leq \frac{B_i}{y_i} = \frac{1}{\sqrt{m_i y_i}},$$

and therefore

$$|d_i| \leq B_i + |c_i|\,|w_i|.$$

Likewise $a_i w_i + b_i = u_i(c_i w_i + d_i)$, so

$$|a_i w_i + b_i| \leq U_i B_i.$$

Taking imaginary parts once more and then using the triangle inequality gives

$$|a_i| \leq \frac{U_i B_i}{y_i}, \qquad |b_i| \leq U_i B_i + |a_i|\,|w_i|.$$

Thus every real embedding of every matrix entry is bounded by an explicit constant. The Minkowski embedding of the chosen coefficient lattice is discrete in a finite-dimensional real vector space, so only finitely many lattice elements satisfy these simultaneous archimedean bounds. Enumerating them and imposing determinant one gives the claimed effective finite set. □

Corollary 7.3 (Finite geometric search for each CM candidate)**.** *After the explicit component normalization of Proposition 7.1 (and any independently chosen level cover), the quotient-membership test for each of the* 12 *strongest maximal-order $d = 3$ candidates reduces to a finite computation. In particular, the infinite union in (6.5) does not require an unbounded search through $\mathrm{SL}_2(\mathcal{O}_L)$: only matrices satisfying the archimedean bounds of Proposition 7.2 can occur.*

The generic finite-search argument requires a specific finite-index model of the curve in X_L, a compact polygon D, and certified bounds m_i, U_i for its normalized component maps. The corrected Schwarz equations in Proposition 12.11 provide one route to such bounds. For the fixed packet considered here, the explicit two-region construction of Lemma I.3.5 and the exact certificates assembled in Theorem I.3.6 complete the geometric decisions.

8 Order-discriminant stratification of the 42 traces

There is a further exact reduction at 43. The field discriminants are

$$\operatorname{disc}(L) = 453789 = 3^3 7^5, \qquad \operatorname{disc}(M) = 205924456521 = 3^6 7^{10}.$$

Hence

$$|\operatorname{disc}(M)| = \operatorname{disc}(L)^2.$$

By the discriminant tower formula, the relative discriminant ideal $\mathfrak{D}_{M/L}$ has norm one and is therefore the unit ideal. In particular, M/L is unramified at all finite primes.

For $\alpha_k = \zeta^k \alpha_0$ and $c_k = \alpha_k + \bar{\alpha}_k$, the relative discriminant of the basis $(1, \alpha_k)$ is

$$\Delta_k = (\alpha_k - \bar{\alpha}_k)^2 = c_k^2 - 4\pi. \tag{8.1}$$

Since the maximal relative discriminant is the unit ideal, $\mathcal{O}_L[\alpha_k] = \mathcal{O}_M$ exactly when $(\Delta_k) = \mathcal{O}_L$.

Theorem 8.1 (Maximal-order trace classes at 43)**.** *Among the* 42 *trace classes of Theorem 4.1, exactly* 16 *generate the maximal CM order* $\mathcal{O}_M$*. With the indexing* $\alpha_k = \zeta^k \alpha_0$*, they are*

$$k \in \{1, 7, 11, 13, 16, 17, 19, 20, 22, 28, 32, 34, 37, 38, 40, 41\}.$$

The absolute norms of the relative order discriminants across all 42 *classes belong to*

$$\{1, 49, 1681, 1849, 2401, 6889, 16129\} = \{1, 7^2, 41^2, 43^2, 49^2, 83^2, 127^2\}.$$

Exactly 12 *of the* 16 *maximal-order classes are also among the* 24 $d = 3$ *arithmetic* $(3, 3)$ *classes, namely*

$$\{1, 11, 16, 17, 19, 20, 22, 32, 37, 38, 40, 41\}.$$

None of the four $d = 7$ *arithmetic* $(3, 3)$ *classes is maximal-order.*

Proof. Equation (8.1) is evaluated exactly in the power basis of L, and its absolute norm is the resultant with the minimal polynomial (I.2.1). The resulting norm list is printed in the computational certificate. Because $\mathfrak{D}_{M/L} = \mathcal{O}_L$, norm one is equivalent to equality of the generated order with $\mathcal{O}_M$. Intersecting the resulting 16 indices with the certified lists in Theorem 5.1 gives the final two assertions. □

Remark 8.2. The maximal-order cases are particularly attractive for the conjugacy repair. The cyclotomic field $M = \mathbb{Q}(\zeta_{42}) = \mathbb{Q}(\zeta_{21})$ has class number one by Masley's Main Theorem [2, p. 179]. This removes the ordinary ideal-class obstruction in $\mathcal{O}_M$; however, one must still track the precise $\mathcal{O}_L$-module, determinant, and SL_2 versus GL_2 equivalence conditions before concluding uniqueness of an admissible Hecke conjugacy class.

Proposition 8.3 (All-trace reduction by embedded order)**.** *The* 42 *traces generate exactly nine embedded orders* $R_k = \mathcal{O}_L[\alpha_k]$ *in the fixed field* M*. Their conductors in* $\mathcal{O}_M$ *are the extensions of the following ideals of* $\mathcal{O}_L$*:*

Conductor in $\mathcal{O}_L$	*Trace indices* k
$\mathcal{O}_L$	$1, 7, 11, 13, 16, 17, 19, 20, 22, 28, 32, 34, 37, 38, 40, 41$
$(7, t-1)$	$0, 3, 6, 9, 15, 18, 21, 24, 27, 30, 36, 39$
$(7, t-1)^2$	$12, 33$
$(41, t-12)$	$14, 35$
$(41, t-22)$	$5, 26$
$(43, t-20)$	$4, 25$
$(43, t-31)$	$8, 29$
$(83, t-36)$	$2, 23$
$(127, t-94)$	$10, 31$

Within each row, an integral invertible affine change of generator gives an $\mathrm{SL}_2(\mathcal{O}_L)$*-equivariant correspondence of all integral actions, preserving their upper-half-plane fixed tuples. For every trace, the maximal-multiplier population is represented by the same sixteen maximal points in Theorem 8.7.*

Proof. Write $\alpha_k = a_k + b_k\zeta$ in $\mathcal{O}_M = \mathcal{O}_L + \mathcal{O}_L\zeta$. Then $R_k = \mathcal{O}_L + b_k\mathcal{O}_M$, and its conductor is $b_k\mathcal{O}_M$: if $A + B\zeta$ and its product with ζ lie in R_k, then both B and $A + tB$ are divisible by b_k, so A is too. For each prime-conductor row, the two exact implementations listed in Section 13 verify $|N_{L/\mathbf{Q}}(b_k)| = p$ and $b_k(r) = 0 \pmod p$ for the displayed root r. Inclusion and equal index give $(b_k) = (p, t - r)$. For the square-conductor row, $b_k/(t-1)^2$ and its inverse are integral units, and $|N_{L/\mathbf{Q}}(t-1)| = 7$. All within-row ratios $e = b_l/b_k$ are checked to be integral units. Thus $\alpha_l = s + e\alpha_k$, with $s = a_l - ea_k \in \mathcal{O}_L$, and $T \mapsto sI + eT$ is the required correspondence. The fixed-point equation is multiplied by the nonzero scalar e at each embedding.

For maximal multiplier, multiplication by α_k in an orientation sector $\delta \in \mathcal{O}_L^\times$ has matrix

$$\begin{pmatrix} a_k & -\delta b_k \\ b_k/\delta & a_k + tb_k \end{pmatrix}.$$

Here $b_k \neq 0$, since otherwise $\pi = a_k^2$ would have square absolute norm. Its upper-half-plane tuple has

$$x = -\delta t/2, \qquad y^2 = \delta^2(4 - t^2)/4, \qquad |w|^2 = \delta^2,$$

independently of k. The maximal Picard group is trivial, and Theorem 11.3 gives the same 32 determinant sectors for every trace. The root identifications in Theorem 8.7 reduce them to the same sixteen representatives. □

Remark 8.4. The two conductor ideals of norm 41 are different embedded ideals. Also the basis change from $(1, \alpha_k)$ to $(1, \alpha_l)$ has determinant e, so companion orientation labels need not be unchanged. This proposition gives no count of all proper modules and no new curve decision; those require the separate class and orbit analyses.

Theorem 8.5 (Exact nonmaximal $d = 7$ conjugacy counts)**.** *For the four $d = 7$ arithmetic $(3, 3)$ trace classes*

$$k \in \{2, 14, 23, 35\},$$

put $R_k = \mathcal{O}_L[\alpha_k] \subset \mathcal{O}_M$. Their conductors are

$$\mathfrak{f}_2 = \mathfrak{f}_{23} = (83, t - 36), \qquad \mathfrak{f}_{14} = \mathfrak{f}_{35} = (41, t - 12).$$

The two conductor primes have residue degree one in L and are inert in the quadratic extension $M/L = L(\sqrt{-7})/L$. The proper ideal class groups and full ideal-class monoids are

k	$N\mathfrak{f}_k$	$\lvert\mathrm{Pic}(R_k)\rvert$	$\lvert\mathrm{ICM}(R_k)\rvert$
$14, 35$	41	1	2
$2, 23$	83	2	3.

Within any fixed totally-positive determinant-orientation sector, refining the corresponding $\mathrm{GL}_2^+(\mathcal{O}_L)$ *similarity classes to determinant one gives*

$$3 \quad (k = 14, 35), \qquad 5 \quad (k = 2, 23)$$

classes. Across all determinant-sign sectors, the full numbers of $\mathrm{SL}_2(\mathcal{O}_L)$ *similarity classes are*

$$96 \quad (k = 14, 35), \qquad 160 \quad (k = 2, 23).$$

Thus the nonmaximal $d = 7$ conjugacy problem is completely finite, but the full oriented enumeration is substantially larger than the single totally-positive sector.

Proof. Since the maximal relative discriminant of M/L is the unit ideal, the order-discriminant formula gives

$$\operatorname{disc}(R_k/\mathcal{O}_L) = \mathfrak{f}_k^2.$$

Exact reduction of $\Delta_k = c_k^2 - 4\pi$ gives

$$N(\Delta_2) = N(\Delta_{23}) = 83^2, \qquad N(\Delta_{14}) = N(\Delta_{35}) = 41^2,$$

and modulo the completely split rational primes one finds that the unique vanishing factors are $t \equiv 36 \pmod{83}$ and $t \equiv 12 \pmod{41}$, respectively. Thus the displayed conductor ideals are exact. Moreover

$$\left(\frac{-7}{41}\right) = \left(\frac{-7}{83}\right) = -1,$$

so the conductor primes are inert in M/L. Therefore locally

$$(\mathcal{O}_M/\mathfrak{f}_k\mathcal{O}_M)^\times/(\mathcal{O}_L/\mathfrak{f}_k)^\times \cong \mathbb{F}_{p^2}^\times/\mathbb{F}_p^\times$$

is cyclic of order $p+1$.

Let $z = \zeta_{42}$, $\rho = z^2 = \zeta_{21}$, and $\varepsilon = 1 - \rho$. Because $\Phi_{21}(1) = 1$, ε is a global unit. At either conductor prime $p \equiv -1 \pmod{42}$, Frobenius is complex conjugation and

$$\varepsilon^{p-1} = \frac{1 - z^{-2}}{1 - z^2} = -z^{-2} = z^{19}.$$

Since 19 is prime to 42, the image of ε in the local quotient has order 42. We must also show that no omitted global unit enlarges this image. For any $\eta \in E_M$, Kronecker's theorem gives $\eta/\bar{\eta} \in \mu(M)$. The map

$$E_M/(\mu(M)E_L) \longrightarrow \mu(M)/\mu(M)^2, \qquad \eta \longmapsto \eta/\bar{\eta},$$

is injective: if $\eta/\bar{\eta} = \omega^2$, then $\eta/\omega \in E_L$. The target has order two, while $\varepsilon/\bar{\varepsilon} = -z^2 = z^{23}$ is not a square in μ_{42}. Consequently

$$E_M = \langle\varepsilon\rangle\mu_{42}E_L.$$

Since $[z] = [\varepsilon]^{22}$ locally, the global-unit image is exactly the order-42 subgroup generated by $[\varepsilon]$. At the conductor-83 prime the element $1 + 3z$ has exact quotient order 84, so it represents the other coset. The standard order class exact sequence and $h(M) = 1$ now give

$$|\operatorname{Pic}(R_{14})| = |\operatorname{Pic}(R_{35})| = 42/42 = 1,$$

$$|\operatorname{Pic}(R_2)| = |\operatorname{Pic}(R_{23})| = 84/42 = 2.$$

For clarity, the ideal-class-monoid completeness can be seen directly in this prime-conductor case. Locally write $A = (\mathcal{O}_L)_\mathfrak{f}$, $B = (\mathcal{O}_M)_\mathfrak{f}$, and $R = A + \mathfrak{f}B$. After scaling a rank-one torsion-free R-lattice I, one has $R \subset I \subset B$. Since B/R is one-dimensional over the residue field of A, necessarily $I = R$ or $I = B$. Away from $\mathfrak{f}$ the order is maximal. Thus the multiplier order is either R_k or $\mathcal{O}_M$, and within either multiplier order the classes form its Picard group. Equivalently, this is the prime-conductor Bass-order decomposition

$$\operatorname{ICM}(R_k) = \operatorname{Pic}(R_k) \sqcup \operatorname{Pic}(\mathcal{O}_M).$$

Since $h(M) = 1$, the displayed ideal-class-monoid counts follow.

For the determinant-one refinement, the integral centralizer of an R_k-multiplier class is $R_k^\times$. In the same local quotient one has $[z] = [\varepsilon]^{22}$. Hence a unit $\varepsilon^n z^m u$, $u \in E_L$, lies in $R_k^\times$ only if

$$n + 22m \equiv 0 \pmod{42},$$

which forces n even. Conversely every square of an E_L-unit is a norm from $R_k^\times$. Hence the norm calculation sharpens to

$$N_{M/L}(R_k^\times) = E_L^2.$$

Because $[E_L^+ : E_L^2] = 2$, every R_k-multiplier class splits into two determinant-one classes inside a fixed totally-positive determinant sector, whereas the $\mathcal{O}_M$-multiplier class has centralizer $\mathcal{O}_M^\times$ and does not split there. This gives $2|\operatorname{Pic}(R_k)| + 1$, namely 3 for conductor 41 and 5 for conductor 83, in such a sector.

For the full $\mathrm{SL}_2(\mathcal{O}_L)$ enumeration one must also retain determinant signs. The maximal-overorder class splits according to

$$E_L/N(\mathcal{O}_M^\times) = E_L/E_L^+,$$

of order 32, while each R_k-multiplier class splits according to

$$E_L/N(R_k^\times) = E_L/E_L^2,$$

of order 64. Hence the total counts are $64|\operatorname{Pic}(R_k)| + 32$: 96 for conductor 41 and 160 for conductor 83. The primary and independent exact certificates construct every module template, all determinant-sign twists, and all 512 trace-labelled representatives. □

Theorem 8.6 (The nonmaximal $d = 3$ classes)**.** *For the twelve selected arithmetic-*$(3, 3)$ *traces*

$$k \in \{0, 4, 8, 12, 15, 18, 21, 25, 29, 33, 36, 39\},$$

there are exactly 1280 *trace-labelled* $\mathrm{SL}_2(\mathcal{O}_L)$ *matrix classes, or* 40 *in one chosen totally positive determinant sector per module class. These are matrix-class counts, not counts of distinct ambient or curve points. The multiplier orders have conductors* $\mathcal{O}_L$, $(7, t-1)$, $(7, t-1)^2$, $(43, t-20)$, *or* $(43, t-31)$, *and each of these orders has trivial Picard group.*

Proof. Proposition 8.3 supplies the conductors. Ten labels have prime conductor and two, $12, 33$, have the square conductor. Every relevant prime splits in M/L. In the conductor unit quotient

$$Q_{\mathfrak{f}} = (\mathcal{O}_M/\mathfrak{f}\mathcal{O}_M)^\times/(\mathcal{O}_L/\mathfrak{f})^\times,$$

use the ratio of the two split coordinates. With $\varepsilon = 1 - z^2$ as above, the following exact table gives generators:

$\mathfrak{f}$	Residue ring	z	$[\varepsilon]$, order
$(7, t-1)$	$\mathbf{F}_7$	3	5, 6
$(7, t-1)^2$	$\mathbf{F}_7[v]/(v^2)$	$3 + 2v$	$5 + 2v$, 42
$(43, t-20)$	$\mathbf{F}_{43}$	30	3, 42
$(43, t-31)$	$\mathbf{F}_{43}$	5	18, 42

In the second row $t = 1+v$: the identity $f(1+v) = 7(1-v) \pmod{v^2}$ proves the displayed residue-ring description. In each row the class of ε generates the entire quotient. The conductor exact sequence and $h(M) = 1$ therefore give trivial Picard groups for these four proper orders. Moreover $[z] = [\varepsilon]^{22}$. The complete relative unit generation $E_M = \langle\varepsilon\rangle\mu_{42}E_L$ proved above shows that $\varepsilon^n z^m w$, $w \in E_L$, is a unit of the proper order precisely when $n + 22m \equiv 0 \pmod{d}$, where $d = 6$ or 42. As $\gcd(22, d) = 2$, its exponent n is even. Consequently the proper-order norm image is exactly E_L^2; the maximal-order image is E_L^+.

For completeness of the module list, let D be the completed local DVR, $B = D \times D$, and $R_e = D + \varpi^e B$, with $e = 1$ or 2. Scale any full rank-one R_e lattice I so that $BI = B$. Both coordinate projections of I are surjective. Its image in the residue plane thus contains a vector with both coordinates nonzero, so I contains a unit of B. Scaling by its inverse gives $R_e \subseteq I \subseteq B$. The quotient $B/R_e \simeq D/(\varpi^e)$ is cyclic; its submodules have preimages exactly $R_j = D + \varpi^j B$, $0 \le j \le e$. Their multiplier rings are distinct. Away from the conductor every lattice is locally principal over the maximal order. Globally every lattice is therefore invertible over its own multiplier order, and its class belongs to that order's Picard group. This proves there are exactly $e+1$ module classes, represented by the overorders themselves, including the intermediate conductor-7 order in the square-conductor case.

Since $h(L) = 1$, each lattice is free of rank two over $\mathcal{O}_L$. The characteristic polynomial is irreducible over L, so integral matrix classes correspond to these rank-one R_k modules. The determinant of a centralizer unit is its relative norm; a module with multiplier S therefore contributes $E_L/N(S^\times)$ determinant-one classes. This gives 32 for maximal multiplier and 64 for each proper multiplier, or 1 and 2 in a fixed positive determinant sector. The totals are $10(32 + 64) + 2(32 + 2 \cdot 64) = 1280$ and $10(1 + 2) + 2(1 + 2 \cdot 2) = 40$. The two exact implementations reconstruct all representatives and check their traces and determinants; completeness and nonconjugacy follow from the module and determinant quotients just proved. □

Theorem 8.7 (The 112 ambient nonmaximal $d = 7$ CM points)**.** *Let $\mathcal{X}_{41}$ and $\mathcal{X}_{83}$ be the sets of $\mathrm{SL}_2(\mathcal{O}_L)$-quotient upper fixed points arising from the trace pairs $\{14, 35\}$ and $\{2, 23\}$, respectively. Then*

$$|\mathcal{X}_{41}| = 48, \qquad |\mathcal{X}_{83}| = 80, \qquad |\mathcal{X}_{41} \cap \mathcal{X}_{83}| = 16,$$

and hence

$$|\mathcal{X}_{41} \cup \mathcal{X}_{83}| = 112.$$

The union has the disjoint multiplier-order decomposition

$$32 \text{ proper-}R_{41} \sqcup 64 \text{ proper-}R_{83} \sqcup 16 \text{ maximal-}\mathcal{O}_M.$$

Thus the 512 objects of Theorem 8.5 are trace-labelled oriented matrix classes, not 512 distinct quotient points.

The normalized A, B, C vertices are three of the 16 maximal-multiplier points. Exact determinant-orientation fingerprints identify two further maximal points with the companion pairs $\{1, 22\}$ and $\{17, 38\}$ excluded in Proposition 11.9. These vertex and residual results alone leave 107 ambient membership decisions: 11 maximal-multiplier and 96 proper-order points. The further exclusions below complete these historical v45 membership decisions in Corollary 9.12.

Proof. Fix one trace $c = c_k$. For every elliptic matrix T, the matrix $\bar{T} = cI - T$ has the same fixed-point equation. The involution $T \mapsto \bar{T}$ is free on $\mathrm{SL}_2(\mathcal{O}_L)$ similarity classes. Indeed, at every real embedding a conjugator from T to $\bar{T}$ must exchange the upper and lower eigenlines, and therefore has negative determinant. More algebraically, every anti-intertwiner is complex conjugation

followed by an element m of the CM centralizer, so its determinant has the totally negative form $-N_{M/L}(m)$. It cannot have determinant one.

Conversely, if two classes give the same quotient point, conjugate them so that they fix the same upper tuple. Their real coefficients make them fix the same lower tuple as well, so both lie in the same quadratic stabilizer algebra. The polynomial $X^2 - cX + \pi$ has only its two roots there. Hence every fixed point fiber has exactly the two classes T and $cI - T$. The 96 and 160 classes therefore give 48 and 80 points for a fixed conductor trace.

The exact identities

$$\alpha_{23} = -\alpha_2, \qquad \alpha_{35} = -\alpha_{14}$$

show that the two traces for each conductor have the same point set. If

$$\gamma_{41} = -1 - 2t + t^2 - t^3 - t^4, \qquad \gamma_{83} = -2 - 2t + 2t^2 - t^4,$$

then exact reduction gives

$$(-174 - 718t - 267t^2 + 733t^3 + 131t^4 - 167t^5)\gamma_{41} + 30\gamma_{83} = 1.$$

Thus the conductor ideals are comaximal. A point supporting both proper orders has multiplier ring containing the order they generate, namely $\mathcal{O}_M$. Conversely every maximal-multiplier point supports all four α_k. The intersection is therefore exactly the maximal stratum, whose 32 matrix classes give 16 points. Halving within each multiplier stratum yields

$$64/2 = 32, \qquad 128/2 = 64, \qquad 32/2 = 16,$$

and proves the 112-point decomposition. Equivalently, each proper point occurs four times in the trace-labelled manifest and each maximal point eight times, giving $32 \cdot 4 + 64 \cdot 4 + 16 \cdot 8 = 512$.

Finally, the integral generators Z_A, B_t, C_t all generate $\mathcal{O}_M$, so direct and conjugate evaluation of every α_k gives the complete two-class fiber over each of the three distinct vertices. In the exact unit-sign basis $(\xi_5, \xi_{11}, \xi_{13}, \xi_{17}, -1, 4 - t^2)$ their maximal-point masks are $7, 15, 0$. The companion bridges for $\{1, 22\}$ and $\{17, 38\}$ have masks 13 and 2. These five masks are distinct, leaving $11 + 96 = 107$ ambient points. Both exact implementations verify every representative, root pairing, and mask. □

9 Historical exclusion batches and their exact scope

The following records retain their original target populations and counts. Only the complete two-region theorem of Paper I supplies the full-curve conclusion for the whole packet. In every statement below that concludes curve exclusion, the packet execution premise of Paper I is inherited; no local summary removes that premise.

Historical first-target computations and their present use. The following subpacket tables retain the original-product P_0 counts, matrix lists and numerical mask digests. Their old checkpoint counts such as 107, 105 and 75 describe that historical progression, not the current undecided population. The full-curve exclusion statements use Theorem I.3.6 in addition to these computations. For the nonprincipal proper-R_{83} module, the old and current arithmetic censuses each describe the entire same 32-point population. The new reflected search excludes all of it. No permutation identifying an individual old numerical mask with a current mask is assumed or needed; no historical digest is relabelled as a reflected digest.

Proposition 9.1 (Two proper-order $d = 7$ exclusions)**.** *In the canonical point manifest of Theorem 8.7, the two nontrivial-R_{83} points represented by the free root pairs* $\{14, 62\}$ *and* $\{30, 46\}$ *do not lie on the normalized McMullen curve. In the historical v45 first-target bookkeeping, this pair reduced* 107 *to* 105*:* 11 *maximal-multiplier and* 94 *proper-order points.*

Proof. Take the index-2 action on the nontrivial conductor-83 ideal lattice used in Theorem 8.7, with determinant-sector mask 14. The absolute norm of its raw lower-left matrix entry is

$$2097130411225081.$$

Thus diagonal unit balancing alone preserves a prohibitively small product of the six coordinate heights. The integral determinant-one inversion

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

replaces it by the negative upper-right entry, of absolute norm $83^2 = 6889$. Put $u = 4 - t^2$ and then apply $H = \mathrm{diag}(u^{-2}, u^2) \in \mathrm{SL}_2(\mathcal{O}_L)$. Both transformations have integral inverses and hence preserve the orbit membership question. For the resulting point the exact squared-height norm is $1/28217344$.

The compact polygon bounds used in Proposition 11.9 are rederived for this source point, using its exact $\mathrm{Re}\, w$, $\mathrm{Im}(w)^2$, and $|w|^2$. Outward rational bounds and the positive trace form give exactly 23923 possible a-entries, 6605 possible c-entries, and 2093 coupled bottom rows (c, d). For $c \neq 0$, the determinant equation forces $b = (ad-1)/c$; exact divisibility and numerator bounds leave 9036 matrices. The separate $c = 0$ branch is empty. Exact cross-multiplied circle signs reject every matrix from at least one coordinate quadrilateral, while retaining boundary equality. Thus there is no polygon survivor.

For the second point, bit 16 changes the orientation unit by -1. If $J = \mathrm{diag}(1, -1)$, exact matrix arithmetic gives $M_{30} = JM_{14}J^{-1}$ for the reduced matrices, hence their fixed points have centers x and $-x$ and common squared height and modulus. Here $\det J = -1$ is totally negative, so J lies in neither $\mathrm{GL}_2^+(\mathcal{O}_L)$ nor $\mathrm{SL}_2(\mathcal{O}_L)$ and does not identify the quotient points. Nevertheless

$$\iota \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}$$

is an involution of $\mathrm{SL}_2(\mathcal{O}_L)$ that bijects their complete bounded matrix sets, including the $c = 0$ sector: for $j(z) = -\overline{z}$ one has $\iota(\gamma)j(z) = j(\gamma z)$. We do not assume that the quadrilateral is reflection-invariant: every reflected matrix is tested separately at the mask-30 point.

The primary rational/SymPy and separate standard-library/Sturm implementations reproduce both empty survivor sets and the respective canonical exclusion digests

`19f1bba6f56200bac45a52810a56669c51824f22a9b3fd410db6a1fc2d69a5a1`,

`2334a40fed0d008d44c2e892871bec20fc04b226a99df899a8927df81595faf6`.

The root involution is xor 48, so $\{14, 62\}$ and $\{30, 46\}$ are disjoint points. Removing them from the 107 points left by Theorem 8.7 gives $11 + 94 = 105$. □

Proposition 9.2 (Two further proper-order $d = 7$ exclusions)**.** *In the canonical point manifest of Theorem 8.7, the two nontrivial-R_{83} points represented by the free root pairs* $\{15, 63\}$ *and* $\{31, 47\}$ *do not lie on the normalized McMullen curve. In the historical v47 first-target bookkeeping, this pair reduced* 105 *to* 103*:* 11 *maximal-multiplier and* 92 *proper-order points.*

Proof. Use the index-2 action on the nontrivial conductor-83 ideal lattice. For each of masks 15 and 31, first apply the integral determinant-one inversion from Proposition 9.1, then conjugate by $\mathrm{diag}(q, q^{-1})$, where q has saturated-unit exponent vector $(1, 1, 1, 0, 0)$ in $(\xi_5, \xi_{11}, \xi_{13}, \xi_{17}, 4 - t^2)$. Both maps have integral inverses. The resulting lower-left entry has absolute norm 83^2, and the exact squared-height norm is $1/28217344$.

The point-specific compactness bounds are rederived with rational Sturm intervals and rounded outward. Complete positive-trace-form enumeration gives 24103 possible a-entries, 6611 possible c-entries, and 2045 coupled bottom rows. Exact determinant solving and coefficient divisibility leave 8278 matrices with $c \neq 0$; the separately enumerated $c = 0$ branch is empty.

For $J = \mathrm{diag}(1, -1)$, the two reduced action matrices satisfy $M_{31} = JM_{15}J^{-1}$. The determinant of J is totally negative, so $J \in \mathrm{GL}_2(\mathcal{O}_L) \setminus \mathrm{GL}_2^+(\mathcal{O}_L)$ and $J \notin \mathrm{SL}_2(\mathcal{O}_L)$: this does not identify the two quotient points. As in Proposition 9.1, the involution

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}$$

bijects their complete bounded determinant-one sets, in both directions and including the $c = 0$ sector. No quadrilateral-reflection symmetry is assumed; all four exact side predicates at all six real embeddings are evaluated separately for both centers. Neither point has a survivor.

The primary rational/SymPy and separate standard-library/Sturm implementations reproduce the two canonical exclusion digests

`66684a7a1aa493a7e37b6b4fc6e97b3663c70e79700724eaca94e170a4d946b4`,

`2fa340baeab5f877f4ea61d9c2e9cc850ee54221d257396b1320dc2c048ef023`.

The v45 root involution is xor 48, so $\{15, 63\}$ and $\{31, 47\}$ are disjoint and are also disjoint from the two fibers in Proposition 9.1. Removing these two further points from the 105 left there gives $11 + 92 = 103$. □

Proposition 9.3 (A third pair of proper-order $d = 7$ exclusions)**.** *In the canonical point manifest of Theorem 8.7, the two nontrivial-R_{83} points represented by the free root pairs* $\{6, 54\}$ *and* $\{22, 38\}$ *do not lie on the normalized McMullen curve. At the historical v49 checkpoint, exactly* 101 *ambient $d = 7$ points remained unclassified for curve membership:* 11 *maximal-multiplier and* 90 *proper-order points.*

Proof. Apply the integral determinant-one inversion of Proposition 9.1 to the index-2 nontrivial conductor-83 ideal action, then conjugate by $\mathrm{diag}(q, q^{-1})$ with $q = \xi_{11}\xi_{13}$. This is the saturated-unit exponent vector $(0, 1, 1, 0, 0)$. Both maps and their inverses are integral. The reduced lower-left entry has absolute norm 83^2, and the exact squared-height norm is $1/28217344$.

Point-specific outward rational bounds, certified from Sturm intervals, give 23937 possible a-entries, 6605 possible c-entries, and 2019 coupled bottom rows. Exact determinant solving and coefficient divisibility leave 8514 matrices with $c \neq 0$; exhaustive enumeration of the separate $c = 0$ branch gives none.

The reduced action matrices satisfy $M_{22} = JM_6J^{-1}$ for $J = \mathrm{diag}(1, -1)$. Since $\det J = -1$ is totally negative, $J \in \mathrm{GL}_2(\mathcal{O}_L) \setminus \mathrm{GL}_2^+(\mathcal{O}_L)$ and $J \notin \mathrm{SL}_2(\mathcal{O}_L)$; it does not identify the quotient points. The involution

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}$$

bijects their complete bounded determinant-one sets in both directions, including $c = 0$, because the exact numerator and denominator norms agree at the reflected centers. No quadrilateral-reflection

symmetry is assumed: all four exact side predicates at all six real embeddings are evaluated separately at both points. Neither test has a survivor.

The primary rational/SymPy and separate standard-library/Sturm implementations reproduce the canonical exclusion digests

`756c33defcade60c47adab07c30af7b5beb7b110e2051e8dfe025055690cb523`,

`5f5c2adaa1366bceab1d2b43a49fffc48c8377320e9d4d522853e1586950e04a`.

The xor-48 fibers $\{6, 54\}$ and $\{22, 38\}$ are disjoint from one another and from the four earlier excluded proper-order fibers. Removing these two further points from the 103 left by Proposition 9.2 gives $11 + 90 = 101$. □

Proposition 9.4 (A fourth pair of proper-order $d = 7$ exclusions)**.** *In the canonical point manifest of Theorem 8.7, the two nontrivial-R_{83} points represented by the free root pairs* $\{1, 49\}$ *and* $\{17, 33\}$ *do not lie on the normalized McMullen curve. At the historical v50 checkpoint, exactly* 99 *ambient* $d = 7$ *points remained unclassified for curve membership:* 11 *maximal-multiplier and* 88 *proper-order points.*

Proof. Apply the integral determinant-one inversion of Proposition 9.1 to the index-2 nontrivial conductor-83 ideal action, then conjugate by $\mathrm{diag}(q, q^{-1})$ with $q = \xi_5\xi_{17}^{-1}(4 - t^2)^{-1}$. This is the saturated-unit exponent vector $(1, 0, 0, -1, -1)$. Both maps and their inverses are integral. The reduced lower-left entry has absolute norm 83^2, and the exact squared-height norm is $1/28217344$.

Point-specific outward rational bounds, certified from Sturm intervals, give 23959 possible a-entries, 6601 possible c-entries, and 2017 coupled bottom rows. Exact determinant solving and coefficient divisibility leave 8748 matrices with $c \neq 0$; exhaustive enumeration of the separate $c = 0$ branch gives none.

The reduced action matrices satisfy $M_{17} = JM_1J^{-1}$ for $J = \mathrm{diag}(1, -1)$. Since $\det J = -1$ is totally negative, $J \in \mathrm{GL}_2(\mathcal{O}_L) \setminus \mathrm{GL}_2^+(\mathcal{O}_L)$ and $J \notin \mathrm{SL}_2(\mathcal{O}_L)$; it does not identify the quotient points. The involution

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}$$

bijects their complete bounded determinant-one sets in both directions, including $c = 0$, because the exact numerator and denominator norms agree at the reflected centers. No quadrilateral-reflection symmetry is assumed: all four exact side predicates at all six real embeddings are evaluated separately at both points. Neither test has a survivor.

The primary rational/SymPy and separate standard-library/Sturm implementations reproduce the canonical exclusion digests

`3d6a732cc36a03fe90933cdfcffd1b9d977e7ce22cb1d7ae1f3926e6dff2e1a8`,

`82b2d957ba62af939e5bb39b0f6bbec06b230094aeee60faa3ae54caf3e2e17b`.

The xor-48 fibers $\{1, 49\}$ and $\{17, 33\}$ are disjoint from one another and from the six earlier excluded proper-order fibers. Removing these two further points from the 101 left by Proposition 9.3 gives $11 + 88 = 99$. □

Proposition 9.5 (A twenty-point proper-order $d = 7$ exclusion batch)**.** *In the canonical point manifest of Theorem 8.7, the twenty nontrivial-R_{83} points with canonical masks*

$$\{0, 16, 2, 18, 3, 19, 4, 20, 5, 21, 8, 24, 9, 25, 10, 26, 11, 27, 12, 28\}$$

do not lie on the normalized McMullen curve. Here the point with mask m is represented by the free root fiber $\{m, m \operatorname{xor} 48\}$. At the historical v51 checkpoint, exactly 79 ambient $d = 7$ points remained unclassified for curve membership: 11 maximal-multiplier and 68 proper-order points.

Proof. For each base mask

$$m \in \{0, 2, 3, 4, 5, 8, 9, 10, 11, 12\},$$

reconstruct the index-2 action on the nontrivial conductor-83 ideal lattice. Apply the integral determinant-one inversion of Proposition 9.1, followed by the point-specific saturated-unit conjugation $\operatorname{diag}(q_m, q_m^{-1})$. Exact inversion in $\mathcal{O}_L$ verifies that q_m and q_m^{-1} are integral, so these reductions and their inverses lie in $\mathrm{SL}_2(\mathcal{O}_L)$. The reduced lower-left entries have absolute norm 83^2, while the exact squared-height and squared-modulus norms are respectively

$$\frac{1}{28217344}, \qquad \frac{2097130411225081}{6889}.$$

For each base point, rational Sturm intervals derive point-specific outward compact bounds. The exact positive trace form and its rational LDL^{T} decomposition enumerate every algebraic integer in the resulting six-place boxes. The complete counts are

m	$\#a$	$\#c$	$\#(c,d)$	$c \neq 0$	$c = 0$
0	24035	6633	2023	8136	0
2	24137	6607	1967	8334	0
3	23993	6597	1937	7794	0
4	24059	6599	2015	8702	10
5	23959	6619	2083	8878	0
8	23967	6613	2021	8750	6
9	24001	6607	2085	9146	0
10	24039	6611	2017	9436	6
11	23973	6617	2013	8360	0
12	24013	6635	2015	8508	4

For $c \neq 0$, the determinant equation forces $b = (ad - 1)/c$, whose integrality and numerator bound are decided exactly. For $c = 0$, it forces $a = d^{-1}$, after which a second exact trace-form enumeration exhausts every bounded b. Thus the last two columns include the entire determinant-one search. At mask 3, the vector $(-7, 0, 0, 0, 0, 0)$ meets the embedding-1 c-entry bound exactly; the closed-box comparison accepts that local equality, while another embedding rejects the vector from the full six-place box.

The bit-16 partner of each base point has center $-x_m$ and the same squared height and modulus. Conjugation by $J = \operatorname{diag}(1, -1)$ induces

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}.$$

Exact numerator and denominator norm identities prove that this is an involutive bijection of the two complete bounded candidate sets, including the nonempty $c = 0$ sectors. But $\det J = -1$ is totally negative, so $J \in \mathrm{GL}_2(\mathcal{O}_L) \setminus \mathrm{GL}_2^+(\mathcal{O}_L)$ and $J \notin \mathrm{SL}_2(\mathcal{O}_L)$; it does not identify the quotient points. No reflection symmetry of the normalized quadrilateral is used. All four exact side predicates at all six embeddings are evaluated separately at each of the twenty centers, and every survivor set is empty.

The primary rational/SymPy and separate standard-library/Sturm paths agree on all twenty first-failure histograms and point digests and on all ten labelled pair aggregates. The primary path additionally freezes the labelled twenty-mask and ten-pair batch digests

`be161e65208adb27deadd69d0d63cbe5a41d500ebd331c02ee7bf22157a47278`,

`17fcca5c094efb163b7eb39636429b5b1b75a56ca6cc4930816742c59607c046`.

The primary path replays the already certified v45 root-pair records; the independent path also reconstructs determinant-one anti-intertwiner bridges for all twenty labels. Hence the twenty xor-48 fibers are pairwise disjoint and disjoint from the eight proper-order fibers excluded earlier. Removing these twenty points from the 99 left by Proposition 9.4 gives $11 + 68 = 79$. □

Proposition 9.6 (The final two proper-order $d = 7$ exclusion pairs)**.** *In the canonical point manifest of Theorem 8.7, none of the four nontrivial-R_{83} points represented by the free root fibers*

$$\{13, 61\}, \quad \{29, 45\}, \quad \{7, 55\}, \quad \{23, 39\}$$

lies on the normalized McMullen curve. The first pair reduces the remainder of Proposition 9.5 to $77 = 11 + 66$*; the second reduces it to*

$$75 = 11 \text{ } \textit{maximal-multiplier} + 64 \text{ } \textit{proper-order}.$$

Proof. For masks 13 and 7, reconstruct the index-2 action on the nontrivial conductor-83 ideal lattice, apply the integral determinant-one inversion of Proposition 9.1, and then conjugate by $\mathrm{diag}(q, q^{-1})$. In the saturated-unit basis $(\xi_5, \xi_{11}, \xi_{13}, \xi_{17}, 4-t^2)$ the exponent vectors are respectively

$$(1, 0, 1, 0, 0), \qquad (0, 0, 0, -1, -2).$$

Exact multiplication in $\mathcal{O}_L$ verifies both units and their integral inverses, so every reduction and its inverse belongs to $\mathrm{SL}_2(\mathcal{O}_L)$. The reduced lower-left entries have absolute norm 83^2.

At each real embedding, rational Sturm isolation supplies intervals $h_i^- \le h_i \le h_i^+$ and $r_i \le r_i^+$ for the height and modulus. Before outward decimal rounding, the certificates form the rational upper endpoints

$$\begin{aligned} \widehat{d}_i &= \mathrm{upper_sqrt}(h_i^+/M_i), & \widehat{c}_i &= \mathrm{upper_sqrt}(1/(M_i h_i^-)), \\ \widehat{n}_i &= U_i \widehat{d}_i, & \widehat{a}_i &= \widehat{n}_i / h_i^-, \\ \widehat{D}_i &= \widehat{d}_i + \widehat{c}_i r_i^+, & \widehat{B}_i &= \widehat{n}_i + \widehat{a}_i r_i^+. \end{aligned}$$

Each stored bound is rounded outward separately from its own hatted endpoint; no formula substitutes already rounded entries into the last two expressions. The positive trace form and its exact rational LDL^{T} decomposition therefore enumerate every algebraic integer in each six-place box. The exact counts are

masks	$\#a$	$\#c$	$\#(c,d)$	$c \neq 0$	$c = 0$
13, 29	23917	6625	2043	8768	0
7, 23	23889	6609	1955	8514	0

where the last two columns count bounded determinant-one candidates. For the second row, the 1955 bottom rows split into 1954 rows with $c \neq 0$ and the single row $(c, d) = (0, 0)$. The latter satisfies $ad - bc = 0$ for every top row and therefore contributes no candidate. This singular sector is part of the complete enumeration, not discarded by division by c.

For each row of the table, bit 16 changes the center x to $-x$ and leaves the squared height and modulus fixed. Conjugation by $J = \mathrm{diag}(-1, 1)$ induces the involution

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}$$

on the complete bounded candidate sets. Since $\det J = -1$ is totally negative,

$$J \in \mathrm{GL}_2(\mathcal{O}_L) \setminus \mathrm{GL}_2^+(\mathcal{O}_L), \qquad J \notin \mathrm{SL}_2(\mathcal{O}_L).$$

Thus this involution is only a search bijection and does not identify the quotient points. No reflection symmetry of the quadrilateral is assumed: all four exact circle-side predicates at all six embeddings are evaluated separately at all four centers. Every survivor set is empty.

The primary rational/SymPy and separate standard-library/Sturm paths agree on all counts, failure histograms, and empty survivor sets. Their point digests are

13	`d66a04fa445505dd16cf2dad98ee6c9c57fe16d90aa4bbeb0b1bf2a1c2cc1a15`
29	`b3ce2ca80967f6acf28bf10d6bb1707f330c768313680c369dd9d6d2cb3349de`
7	`b653b69bf5888c06df2ee9a20e5f6fd8d1d06277de0b862933ba5a4bf8811ae5`
23	`392b61bb9ec4dcfd6720de02184032cfe211f9094b3e5ed1dac5224be84f8d52`

with pair aggregates

`20ca8ce844583d40514ca6f65db26a302957b057fb0fd23c057cb7e52dab2e68`,

`aba4002201230fea6eb9e127aa6fa38a04e63194d3970adc32c95a1a87386dab`.

The xor-48 fibers displayed in the statement are mutually disjoint and are disjoint from all fibers excluded through Proposition 9.5. Removing the first two points from 79 gives 77, and removing the last two gives 75. □

Corollary 9.7 (Closure of the nontrivial-R_{83} stratum)**.** *The exclusions in Propositions 9.1–9.6 cover the old canonical masks* $\{0, 1, \ldots, 31\}$ *exactly once. Their original-product emptiness and the complete reflected-population exclusion in Theorem* I.3.6 *exclude all* 32 *points of this stratum. This uses equality of the full oriented populations, not a correspondence between individual numerical mask labels. At the historical v53 first-target checkpoint the remaining* 75 *ambient* $d = 7$ *candidates were exactly* 11 *maximal-multiplier points,* 32 *proper-*R_{41} *points, and* 32 *principal-Picard-class proper-*R_{83} *points.*

Theorem 9.8 (Closure of the principal-Picard proper-R_{83} stratum)**.** *None of the* 32 *ambient quotient points whose multiplier module is in the principal Picard class of the proper order* R_{83} *lies on the normalized McMullen quadrilateral in the trace-polarized* $\mathrm{SL}_2(\mathcal{O}_L)$ *quotient.*

Proof. The v45 ambient classification represents the principal class by the regular lattice $R_{83} = \mathcal{O}_L[\alpha_2]$ with basis $(1, \alpha_2)$. Its 64 determinant sectors are indexed by masks $0, \ldots, 63$. The exact free-root involution is $m \mapsto m \,\texttt{xor}\, 16$: an integral determinant-one bridge carries $\mathrm{Tr}(\alpha_2)I - T_m$ to $T_{m\texttt{xor}16}$. Hence the 32 disjoint root fibers are represented exactly once by

$$\mathcal{M} = \{0, \ldots, 15, 32, \ldots, 47\}.$$

The anti-intertwiner used to compare candidate sets has totally negative determinant. It lies in $\mathrm{GL}_2(\mathcal{O}_L)$ but not in $\mathrm{GL}_2^+(\mathcal{O}_L)$ or $\mathrm{SL}_2(\mathcal{O}_L)$, so it is not used to identify quotient points. The identity $\alpha_{23} = -\alpha_2$ gives the same point set for the second conductor-83 trace.

For each $m \in \mathcal{M}$, the certificate constructs and exactly verifies a cyclotomic unit q_m and applies $\mathrm{diag}(q_m, q_m^{-1}) \in \mathrm{SL}_2(\mathcal{O}_L)$. The six exact fixed-point heights then give outward rational bounds. Before rounding, these imply simultaneous bounds for c, d from the lower-half contribution to the trace-LDL^{T} form and, after c, d are fixed, for a, b from the upper-half contribution. Enumerating the resulting finite trace ellipsoids is therefore complete.

The determinant equation $ad - bc = 1$ is solved in both branches. If $c \neq 0$, the affine solution set is obtained from one exact solution by $(a, b) = (a_0 + nc, b_0 + nd)$, and the exact archimedean intervals give all integers n. If $c = 0$, the code separately imposes $ad = 1$, integrality of $a = d^{-1}$, and the complete bounded list of b; this essential branch contributes 49,232 candidates. An image in the original product P_0 must satisfy every one of its four closed circle-side inequalities in all six embeddings, so a failed exact side sign excludes that candidate from P_0. The reflected-product calculation and Lemma I.3.5, as assembled in Theorem I.3.6, supply the full-curve conclusion.

The primary quotient-field/SymPy implementation and a separate standard-library/Sturm implementation independently reconstruct the order, root bridges, reductions, bounds, determinant branches, and polygon signs. For the 32 centers they agree on 62,262 bounded bottom rows, 219,840 candidates with $c \neq 0$, 49,232 with $c = 0$, and hence 269,072 candidates in total. Every survivor set is empty; all per-center counts, first-failure histograms, and classification digests agree, with order-sensitive batch digest

`7a2f2f6f71efb74627f4f891f108b2548a1a9b2a8584cea684eaa630c6d90d90`.

The exact sources are listed in Section 13. □

Corollary 9.9 (Closure of the proper-R_{83} locus)**.** *Corollary 9.7 and Theorem 9.8 exhaust and exclude all* 64 *proper-R_{83} ambient points. These results and the five earlier maximal decisions settle* 69 *of the* 112 *ambient $d = 7$ points. Theorem 9.10 below adds the remaining eleven maximal exclusions; Theorem 9.11 then excludes all* 32 *proper-R_{41} points, completing the* 112*-point census.*

Here "principal Picard class" means the identity proper-ideal class of R_{83}; it is unrelated to principal polarization. No new CM-type, Hodge, or polarization assertion is made in Theorem 9.8.

Theorem 9.10 (All maximal-multiplier orientations)**.** *In the fixed marked extension M/L, the maximal-multiplier strata of all* 42 *corrected norm-π traces have exactly sixteen ambient quotient points. Exactly three lie on the normalized McMullen curve: A, B, C. The other thirteen are excluded. This statement concerns maximal multiplier order, including its occurrences under non-maximal generated orders; it makes no proper-order curve-membership assertion.*

Proof. Proposition 8.3 identifies the same maximal-multiplier population across all 42 traces; Theorem 8.7 supplies its sixteen-point census. The exact vertex realizations identify masks $0, 7, 15$ with C, A, B, respectively. We exclude the thirteen other masks by checking both targets of Lemma I.3.5. First consider the original product P_0. Its two historical companion calculations concern masks $2, 13$; the label transfer is justified below. The eleven remaining original-product searches use masks

$$1, 3, 4, 5, 6, 8, 9, 10, 11, 12, 14.$$

Use the maximal action of $\alpha_2 = a + bz$, where $a = -2 - t + 3t^2 - t^4$ and $b = -1 + 4t + t^2 - 4t^3 + t^5$. For the established sign representatives δ_m put

$$T_m = \begin{pmatrix} a & -\delta_m b \\ b/\delta_m & a + tb \end{pmatrix}.$$

The free two-root fibers pair m with $m\mathrm{xor}16$. The integral determinant-one matrix $R_m = \left(\begin{smallmatrix} -1 & -\delta_m t \\ 0 & -1 \end{smallmatrix}\right)$ satisfies $R_m(\mathrm{tr}(T_m)I - T_m) = T_{m\mathrm{xor}16} R_m$. Thus no negative-determinant reflection is used to identify quotient points.

Balance by $\mathrm{diag}(q, q^{-1})$, with $q = 1$ for masks $1, 4, 8, 9, 10$, $q = \xi_{11}^{-1}$ for $3, 6$, $q = \xi_5^{-1}$ for $5, 11$, $q = \xi_{17}^{-1}$ for 12, and $q = \xi_{13}^{-1}$ for 14. Here

$$\xi_5 = t^4 - 3t^2 + 1, \quad \xi_{11} = t^3 + t^2 - 2t - 1, \quad \xi_{13} = t^5 - 4t^3 + 3t + 1, \quad \xi_{17} = t^3 - 2t.$$

All these units and their inverses are integral. With $\eta = \delta_m q^2$, the balanced fixed tuple has exact invariants

$$x = -\eta t/2, \qquad h^2 = \eta^2(4 - t^2)/4, \qquad |w|^2 = \eta^2.$$

For each real embedding, use the same certified target-polygon bounds $\mathrm{Im}(\gamma w) \geq M > 0$, $|\gamma w| \leq U$ as in the preceding exclusions. For $\gamma = \left(\begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix}\right)$ of determinant one,

$$|c'w + d'|^2 \leq h/M, \qquad |a'w + b'| \leq U\sqrt{h/M}.$$

Writing certified rational bounds $h^- \leq h \leq h^+$ and $|w| \leq r^+$, these imply outward bounds

$$|c'| \leq \sqrt{1/(Mh^-)}, \quad |a'| \leq U\sqrt{h^+/M}/h^-, \quad |d'| \leq \sqrt{h^+/M} + |c'|r^+, \quad |b'| \leq U\sqrt{h^+/M} + |a'|r^+.$$

Each bound is rounded outward from its own rational pre-rounding endpoint. The positive trace form gives finite complete containing ellipsoids for a' and c'. For each c', the denominator inequality gives a centered ellipsoid for d'. If $c' \neq 0$, the only possible top-right entry is $b' = (a'd' - 1)/c'$; integrality and all numerator bounds are tested exactly. If $c' = 0$, impose $a'd' = 1$ and enumerate every b' in the centered numerator ellipsoid. Thus both determinant branches are exhaustive.

Both implementations independently reconstruct these bounds and enumerate all candidates, retaining boundary equalities. The final predicates are necessary polygon-side inequalities in all six embeddings; rational intervals have exact algebraic sign fallback. Their full results agree:

Mask	Bottom rows	$c' \neq 0$ candidates	$c' = 0$ candidates
1	1975	5280	0
3	1891	6996	144
4	2089	7736	136
5	1975	6748	140
6	2433	8540	112
8	1891	7392	156
9	2177	10840	212
10	2017	8956	168
11	1933	8972	148
12	2143	9796	32
14	2017	7004	36
Total	22541	88260	1284

All 89,544 candidates fail a necessary polygon predicate. The two portable paths agree on every candidate digest and failure histogram, with ordered classification digest

`beff7e2462c7f461f1990c6297381b49c8de2895b59e2be3baa98f0b8d869e2c`.

The full replay covers all eleven masks in both paths. The named exact certificates are listed in Section 13. At this stage the conclusion is avoidance of P_0 for these eleven masks.

For masks 2 and 13, the historical companion calculations for $k = 17, 38$ and $k = 1, 22$, respectively, give 9812 and 9072 candidates per companion, with none in P_0. To transfer these statements, write $\alpha_k = a_k + b_k z$ and use

$$J_k = \begin{pmatrix} 1 & a_k \\ 0 & b_k \end{pmatrix}.$$

It intertwines the companion action and multiplication on $(1, z)$. Here $\mathcal{O}_L[\alpha_k] = \mathcal{O}_M$, so b_k is an integral unit. The exact orientation fingerprints select the stated masks and make $\det(D_{\delta_m} J_k)$ a totally positive unit, where $D_{\delta_m} = \operatorname{diag}(\delta_m, 1)$. Theorem 11.3 supplies a CM unit whose multiplication matrix in the companion centralizer has inverse determinant. Composing with it gives an integral determinant-one intertwiner. Changing the nonreal generator of this fixed embedded CM action preserves its unique upper fixed point at every real place. Thus these are the required labelled P_0 statements, without identifying $\mathrm{GL}_2(\mathcal{O}_L)$ with $\mathrm{SL}_2(\mathcal{O}_L)$.

For the second target set $R = A_t B_t - B_t A_t$. The two exact target implementations verify that $\operatorname{tr}(R) = 0$, that $R^2 = -\det(R) I$, and that $\det(R)$ is negative at every real embedding, with absolute norm 21952. The anti-Mobius action

$$r_i(w) = \frac{\sigma_i(R_{11})\overline{w} + \sigma_i(R_{12})}{\sigma_i(R_{21})\overline{w} + \sigma_i(R_{22})}$$

fixes A_i and B_i and is the reflection in their geodesic. It transforms only the target. The certificates reconstruct the four reflected vertices and all closed side inequalities independently. Valid rational lower height and upper modulus bounds for $r_i(Q_i)$, in the fixed embedding order, are

$$10^5(M_i) = (1368, 23817, 14522, 22806, 10088, 5105),$$

$$10^5(U_i) = (197767, 354283, 28993, 254603, 165248, 191115).$$

The exact vertex comparisons imply the bounds on the entire convex polygon: the horoball $\operatorname{Im}(w) \geq M_i$ and the geodesic half-plane $|w| \leq U_i$ are hyperbolically convex. The preceding determinant-one bounds therefore give complete finite ellipsoids for this target too. Certified unit changes of enumeration coordinates are integral bijections, and every candidate is checked against the original bounds. Both determinant branches and all boundary equalities are retained.

For each of the thirteen labels the source module, sign sector and determinant-one balancing are reconstructed from its frozen exact plan. The two implementations give the following complete reflected-target counts, with zero survivors in every row:

Mask	Bottom rows	$c' \neq 0$ candidates	$c' = 0$ candidates
1	3865	83400	1332
2	4277	91720	1704
3	3693	75752	1300
4	3655	81272	776
5	3945	81240	780
6	3861	89948	772
8	3853	88696	372
9	3991	97736	1800
10	3819	76348	280
11	3945	82188	708
12	4143	103792	1376
13	4067	83148	548
14	4021	66496	440
Total	51135	1101736	12188

In particular all 1,113,924 reflected candidates per path fail a necessary side condition. The canonical preparation, entry lists, bottom rows, both candidate branches and complete classifications agree between the separate implementations. The retained executions, their portable-source correspondence and the exact per-label comparison data are identified in Section 13. These data certify complete searches; comparison expectations do not define the search domain. We now have avoidance of both P_0 and P_1 for each of the thirteen labelled points. Lemma I.3.5 proves their full-curve exclusion. Together with the three distinct positive realizations and the arithmetic census, this proves the theorem. □

Theorem 9.11 (Exclusion of the complete proper-R_{41} stratum)**.** *In the fixed marked extension M/L, none of the* 32 *ambient points with proper multiplier order*

$$R_{41} = \mathcal{O}_L + (41, t-12)\mathcal{O}_M$$

lies on the normalized McMullen curve in the trace-polarized $\mathrm{SL}_2(\mathcal{O}_L)$ *quotient. These are the complete proper-order points arising from traces $k = 14, 35$ in Theorem 8.7. The distinct embedded order of conductor $(41, t-22)$ is not included in this exclusion.*

Proof. Theorem 8.5 gives $\mathrm{Pic}(R_{41}) = 1$ and $N_{M/L}(R_{41}^{\times}) = E_L^2$. Thus the regular module R_{41} supplies exactly 64 matrix classes up to $\mathrm{SL}_2(\mathcal{O}_L)$ conjugacy, indexed by the full unit-square masks $0, \ldots, 63$. Write

$$\tau = c_{14} = t^5 - 7t^3 - t^2 + 11t + 2, \qquad T_m = \begin{pmatrix} 0 & -\delta_m \pi \\ \delta_m^{-1} & \tau \end{pmatrix},$$

where δ_m is the binary product in the ordered basis $(\xi_5, \xi_{11}, \xi_{13}, \xi_{17}, -1, 4-t^2)$. This is the regular basis $(1, \alpha_{14})$, followed by its orientation twist. The integral determinant-one matrix

$$R_m = \begin{pmatrix} -1 & -\delta_m \tau \\ 0 & -1 \end{pmatrix}$$

satisfies

$$R_m(\tau I - T_m) = T_{m\mathrm{xor}16} R_m.$$

By Lemma I.3.1, the two root classes are precisely the fiber over a quotient point. Hence the canonical masks

$$\mathcal{M}_{41} = \{0, \dots, 15, 32, \dots, 47\}$$

represent all 32 disjoint pairs $\{m, m \operatorname{xor} 16\}$ exactly once. The bridge compares the conjugate root action with the opposite mask; it does not identify actions by a negative-determinant reflection. The identity $\alpha_{35} = -\alpha_{14}$ gives the same point population for the second trace.

For each canonical mask, use the explicit real unit q_m in the frozen witness table and conjugate by $\operatorname{diag}(q_m, q_m^{-1}) \in \mathrm{SL}_2(\mathcal{O}_L)$. Both exact paths check integrality of the unit and its inverse and the determinant-one transformation. With $\eta = \delta_m q_m^2$, the balanced upper tuple $w = x + ih$, $h > 0$, has exact invariants

$$x = -\eta\tau/2, \qquad h^2 = \eta^2(4\pi - \tau^2)/4, \qquad |w|^2 = \eta^2\pi.$$

These identities, total positivity of the squared heights, and the fixed ordering of the six real embeddings are checked separately in both implementations.

Here is the finite-domain argument, with the same certified target bounds as in Theorem 9.10. At each embedding, a possible image in the normalized quadrilateral has height at least $M > 0$ and modulus at most U. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ of determinant one,

$$|cw + d|^2 \le h/M, \qquad |aw + b| \le U\sqrt{h/M}.$$

Certified rational intervals $h^- \le h \le h^+$ and $|w| \le r^+$ therefore give upper bounds

$$\widehat{D}_0 \ge \sqrt{h^+/M}, \quad \widehat{C} \ge \sqrt{1/(Mh^-)}, \quad \widehat{N}_0 = U\widehat{D}_0, \quad \widehat{A} = \widehat{N}_0/h^-.$$

The remaining endpoints are $\widehat{D} = \widehat{D}_0 + \widehat{C}r^+$ and $\widehat{B} = \widehat{N}_0 + \widehat{A}r^+$. Each final entry bound is rounded outward from its own pre-rounding endpoint. An integral entry v in the six boxes satisfies $\mathrm{Tr}_{L/\mathbf{Q}}(v^2) \le \sum_i R_i^2$. The positive rational trace form consequently gives finite complete containing ellipsoids, enumerated by exact LDL^{T} recursion.

First enumerate the complete a and c lists. For each c, the centered denominator ellipsoid around $-cx$, followed by its exact six-place inequalities, gives all necessary d. For $c \neq 0$, each a and bottom row has the unique possible $b = (ad - 1)/c$; test its integrality and all numerator bounds. For $c = 0$, reject $d = 0$, impose the unique $a = d^{-1}$, test its integrality and a bounds, and enumerate every b in the centered numerator ellipsoid. Thus the zero branch is included in full. All boundary equalities remain admissible.

The primary quotient-field path `certificates/d7_r41_v63.py` and the separate standard-library/Sturm path `independent/d7_r41_sturm_v63.py` perform these searches and the necessary polygon-side tests in all six embeddings. Their witnesses and exact per-center expectations are retained in `certificates/d7_r41_expected_v63.py`. The final signs use rational intervals with exact algebraic fallback; uncertainty is never accepted as an exclusion. Summed over the 32 centers, each implementation obtains

Bounded bottom rows	62,474
Candidates with $c \neq 0$	231,744
Candidates with $c = 0$	33,468
All determinant-one candidates	265,212
Survivors	0.

Every per-center count, candidate digest, zero-branch audit, first-failure histogram and classification digest agrees. The ordered classification digest is

`d3e983df4d9ff3d9d3ebf498df08345b701eb95a5a0edc4a0b5654a88fb4ba06`.

The complete portable replay covers all 32 masks in each path. These empty sets prove avoidance of the original product P_0. The full reflected 32-point search, with fresh target bounds, is part of Theorem I.3.6. Together with Lemma I.3.5, it proves the stated curve exclusion. □

Corollary 9.12 (Closure of the nonmaximal $d = 7$ geometric packet)**.** *The* 112 *distinct ambient points attached to the four nonmaximal $d = 7$ traces $k = 2, 14, 23, 35$ contain exactly three points of the normalized McMullen curve: A, B, C. The other* 109 *are excluded. Thus this complete $d = 7$ population is classified in the fixed marked, trace-polarized $\mathrm{SL}_2(\mathcal{O}_L)$ model. Its complement in the full arithmetic census of Theorem* I.3.3 *has* 256 *points.*

Proof. The intrinsic multiplier decomposition of Theorem 8.7 has 16 maximal, 64 proper-R_{83} and 32 proper-R_{41} points. Theorem 9.10 gives the three maximal hits A, B, C and thirteen maximal exclusions. Corollary 9.9 excludes all 64 proper-R_{83} points, and Theorem 9.11 excludes the remaining 32. Hence there are $13 + 64 + 32 = 109$ exclusions and three hits. □

Remark 9.13. This closure adds no positive curve point. The previously proved geometric CM types and Hodge conclusions at A, B, C retain their existing scope. "Principal Picard class" for R_{41} refers to an ideal class and not a principal polarization. The result concerns the normalized trace-polarized quotient in the specified marked CM extension; it neither classifies other CM fields nor identifies the standard fibers with Mostaed's principally polarized locus.

Seven pointwise exclusions beyond the $d = 7$ population. The full arithmetic census also permits individual geometric decisions outside the complete population of Corollary 9.12. The following result concerns seven specified points, not their whole multiplier or Picard strata.

Proposition 9.14 (Seven further pointwise exclusions)**.** *In the fixed marked extension and normalized trace-polarized $\mathrm{SL}_2(\mathcal{O}_L)$ quotient, the seven points in the following table do not lie on McMullen's curve. The labels are those of the canonical point manifest of Theorem* I.3.3*; all have orientation mask zero. The last three columns count the complete bounded matrices before the original-product closed polygon tests, in either exact implementation. These counts are retained historical P_0 data.*

Point label	k	*Multiplier conductor*	$c \neq 0$	$c = 0$	*Total*
`R127:pic0:00`	10	$(127, t - 94)$	6572	2304	8876
`R43_20:pic0:00`	4	$(43, t - 20)$	7704	952	8656
`R7sq:pic0:00`	12	$(7, t - 1)^2$	7084	1320	8404
`R41_22:pic0:00`	5	$(41, t - 22)$	7148	1616	8764
`R43_31:pic0:00`	8	$(43, t - 31)$	8556	1436	9992
`R7:pic0:00`	0	$(7, t - 1)$	8840	588	9428
`R127:pic1:00`	10	$(127, t - 94)$	8488	0	8488.

The survivor count is zero in every row. Here `pic0` *means the principal Picard module class, not a principal polarization;* `pic1` *is the nonprincipal class represented at conductor $(127, t - 94)$ by J_9 in the proof of Theorem* I.3.3*.*

Proof. Write $\alpha_k = A_k + B_k z$ as in the arithmetic census. For the six principal rows use the ordered basis $(1, B_k z)$, and for the last row use $(1+9z, B_{10} z)$. Put $s = 0$ or 9 accordingly and $n = 1+st+s^2$. Multiplication by α_k has the integral matrix

$$T_{k,s} = \begin{pmatrix} A_k - sB_k & -B_k^2 \\ n & A_k + (t+s)B_k \end{pmatrix}.$$

The two paths independently reconstruct this action, its characteristic polynomial and its upper fixed tuple $w = x + ih$. In the fixed ordering of the six real embeddings its exact invariants are

$$x = -\frac{B_k(t+2s)}{2n}, \qquad h^2 = \frac{B_k^2(4-t^2)}{4n^2}, \qquad |w|^2 = \frac{B_k^2}{n}.$$

All squared heights are positive. The conductor, module and root-fiber identifications are those of Theorem I.3.3. In particular the last point has root partner $(J_{100}, 16)$; this is the other root representative of the same point, not an additional exclusion.

The portable witness data specify a balancing transformation $G \in \mathrm{SL}_2(\mathcal{O}_L)$, consisting of a diagonal unit transformation, with inversion first when prescribed. Both paths verify its integral inverse and reconstruct the transported point. If $p' = Gp$, an integral determinant-one representative H at p' corresponds to HG at p. Thus this change of representative preserves the exact orbit question.

Apply the finite-domain argument of Theorem 9.11 to the transported tuple. It uses only positive source heights, determinant one and the certified target height/modulus bounds, and therefore applies to the nonprincipal row as well. Its outward rational bounds and positive trace form contain every possible entry in finite ellipsoids. The exact unit changes of enumeration coordinates are integral bijections: every enumerated entry is mapped back before the original six-place bounds are tested.

For clarity, write an original shifted bound at embedding i as $|v_i + q_i x_i|^2 + q_i^2 h_i^2 \le L_i^2$. For an integral unit u, choose certified rational $B_i \ge |u_i| L_i$ and set $y = uv$. Then every admissible v satisfies

$$\mathrm{Tr}_{L/\mathbf{Q}}((y + uqx)^2) \le \sum_i B_i^2 - \mathrm{Tr}_{L/\mathbf{Q}}((uq)^2 h^2).$$

This is the searched ellipsoid centered at $-uqx$; $q = 0$ gives the entry-box searches, $q = c$ the denominator search, and $q = a$ the zero-branch numerator search. Multiplication by u^{-1} recovers every original integral entry, regardless of the embedding signs of u. Strictly negative remaining radii give empty domains; zero radii and boundary equality are retained.

For every admitted c, the denominator enumeration finds all admitted d. If $c \neq 0$, each admitted a has the unique possible completion $b = (ad-1)/c$; integrality and the original numerator inequalities are tested. If $c = 0$, require $d \neq 0$ and $a = d^{-1} \in \mathcal{O}_L$, check the a bounds and enumerate the complete remaining b domain. Hence the zero branch is included even when it returns no matrix.

For the last row the portable computation also rejects nonprimitive bottom rows: $(c, d) \neq \mathcal{O}_L$ precludes $ad - bc = 1$. The ideal index is computed afresh; a unit c directly certifies primitivity. On every remaining nonzero bottom row it verifies integer identities

$$M_c R = D M_d, \qquad M_c v = -D e_0, \qquad D > 0,$$

where M_c, M_d are multiplication matrices in the integral power basis and e_0 represents 1. These identities certify the complete affine formula $b = (Ra + v)/D$ before scanning the entire original a list. Simultaneous negation preserves determinant, action and every original bound. Both matrix signs are restored before canonical sorting and polygon classification; the retained matrices receive exact determinant checks. These reductions therefore omit no bounded candidate.

The six principal rows give 54,120 complete candidates per path. The last row freshly gives 29,555 possible a entries, 8,167 possible c entries, 2,005 bottom rows and 8,488 complete candidates. The primary exact algebra/trace-ellipsoid path and the separate integer/Sturm path agree on the reconstructed preparation, entry lists, bottom rows, both candidate branches and complete polygon

classifications. The named portable certificates and comparison data are listed in Section 13. Comparison targets do not define or truncate the search; the full success status requires recomputed counts and all eight canonical artifact hashes to agree.

Every candidate fails a necessary closed polygon-side condition in at least one embedding, with exact algebraic signs or certified rational intervals. This proves avoidance of P_0 for these seven labels. Their separate reflected searches are included in the full 365-label certificate of Theorem I.3.6; Lemma I.3.5 then proves the seven curve exclusions. □

Corollary 9.15 (Historical subpacket coverage)**.** *The complete $d = 7$ population and the seven additional labels above form a* 119*-point subset of the* 368*-point arithmetic census: the three vertices and* 116 *exclusions. Its complement has* 249 *points, all excluded by Theorem* I.3.6*. The figures* 119 *and* 249 *describe the historical division of the first-target work, not a current coverage total or an unresolved set.*

Proof. The 112-point $d = 7$ census and the seven distinct additional points are disjoint. Outside this subset the four proper $d = 3$ strata contribute $128 - 4 = 124$ points, conductor $(41, t - 22)$ contributes $32 - 1 = 31$, and conductor $(127, t - 94)$ contributes $96 - 2 = 94$. Their sum is 249. Principal controls are not counted a second time, and a paired root is not a second quotient point. The first-target coverages 116 and 249 are disjoint and exhaust all 365 nonvertices. □

Remark 9.16. These seven negative orbit decisions add no positive geometric CM type, K-signature, Hermitian discriminant or Hodge-algebra assertion. They neither exclude a whole remaining conductor or Picard stratum nor identify any standard fiber with a principally polarized model or with Mostaed's named principally polarized locus. The geometric CM and Hodge conclusions at A, B, C are unchanged.

10 Individual self-dual models and lost automorphisms

These constructions establish the local obstructions and special-fiber alternatives. Paper I uses one common nonfree subgroup for its global family; an arbitrary individual choice is not silently substituted for it.

Proposition 10.1 (Principally polarized model at the C-vertex)**.** *The normalized C-vertex admits an $\mathcal{O}_L$- and $\mathcal{O}_K$-stable self-dual overlattice of index*

$$\operatorname{disc}(L) = 453789$$

for $K = \mathbb{Q}(\sqrt{-3})$. Hence it is $\mathcal{O}_L$-linearly isogenous of degree 453789 *to a principally polarized sixfold A_C^{pp} with the same rational CM Hodge structure, signature, discriminant, and Hodge-conjecture conclusion.*

Proof. Write $\Sigma_C = (I + 2C_t^{14})^{\mathsf{T}}$. On the discriminant module

$$\Lambda^{\#}/\Lambda \cong (\mathcal{O}_L/\mathfrak{p}_3)^2 \oplus (\mathcal{O}_L/\mathfrak{p}_7^5)^2$$

one has $\Sigma_C \equiv 0 \pmod{\mathfrak{p}_3}$. Choose any residue line on that part. On the $\mathfrak{p}_7^5$-part, $\Sigma_C^2 = 4I$ and $\operatorname{tr}(\Sigma_C) = 0$, so the idempotents

$$\frac{\Sigma_C + 2I}{4}, \qquad \frac{2I - \Sigma_C}{4}$$

give rank-one eigensummands. Either is isotropic because $\Sigma_C^{\dagger} = -\Sigma_C$. The CRT product has order $3^3 7^5$ and is therefore maximal isotropic. The same argument as for A and B proves $\mathcal{O}_K$-stability, self-duality, preservation of all rational Hodge data. □

Corollary 10.2 (The principal C-model loses the full vertex automorphism)**.** *No principally polarized self-dual overlattice of the standard lattice can retain the full integral automorphism acting as C_t^{T} on homology.*

Proof. Stability under C_t^{T} and its inverse implies stability under tI, so the order-27 subgroup is an $\mathbb{F}_{27}$-line. Modulo $\mathfrak{p}_3$, the characteristic discriminant of C_t^{T} is t^2-1, and exact arithmetic in $\mathbb{F}_{27}=\mathbb{F}_3[t]/(t^3-t^2+t+1)$ gives

$$(t^2-1)^{13}=-1.$$

Thus the characteristic polynomial is irreducible over $\mathbb{F}_{27}$ and C_t has no invariant residue-field line. □

Proposition 10.3 (Principally polarized model at the A-vertex)**.** *Let $\Lambda=\mathcal{O}_L^2$ be the homology lattice at the normalized A-vertex and $\Lambda^{\#}=\mathfrak{D}_L^{-1}\Lambda$ its dual for E_1. There is an $\mathcal{O}_L$- and $\mathcal{O}_K$-stable maximal isotropic subgroup*

$$H_A\subset\Lambda^{\#}/\Lambda,\qquad K=\mathbb{Q}(\sqrt{-3}),$$

whose inverse image Λ_A^{pp} is self-dual for E_1. Hence

$$[\Lambda_A^{\mathrm{pp}}:\Lambda]=\mathrm{disc}(L)=453789$$

and the normalized A-vertex sixfold is $\mathcal{O}_L$-linearly isogenous of degree 453789 to a principally polarized abelian sixfold A_A^{pp}. The rational CM Hodge structure, the K-signature $(3,3)$, the Hermitian discriminant class -1, and the divisor-generation of the Hodge algebra are unchanged. In particular the Hodge conjecture holds for A_A^{pp} in every codimension.

Proof. As in Proposition I.4.3,

$$\Lambda^{\#}/\Lambda\cong(\mathcal{O}_L/\mathfrak{p}_3)^2\oplus(\mathcal{O}_L/\mathfrak{p}_7^5)^2,\qquad\mathfrak{D}_L=\mathfrak{p}_3\mathfrak{p}_7^5.$$

Let Σ_A be the integral operator representing $\sqrt{-3}$ in the proof of Theorem I.5.14. Exact reduction gives

$$\Sigma_A\equiv 0\pmod{\mathfrak{p}_3}.$$

Thus any one-dimensional $\mathbb{F}_{27}$-subspace of the $\mathfrak{p}_3$-part is Σ_A-stable; fix one and call it $H_{A,3}$.

On the $\mathfrak{p}_7^5$-part one has $\Sigma_A^2=-3I=4I$ and $\mathrm{tr}(\Sigma_A)=0$. Since 4 is a unit, the idempotents

$$e_{A,+}=\frac{\Sigma_A+2I}{4},\qquad e_{A,-}=\frac{2I-\Sigma_A}{4}$$

split the rank-two local module into two rank-one direct summands. Put $H_{A,7}=\mathrm{im}(e_{A,+})$. The relation $\Sigma_A^{\dagger}=-\Sigma_A$ gives, for $v,w\in H_{A,7}$,

$$0=E_1(\Sigma_Av,w)+E_1(v,\Sigma_Aw)=4E_1(v,w),$$

so $H_{A,7}$ is isotropic. Therefore

$$H_A=H_{A,3}\oplus H_{A,7}$$

is isotropic and has order

$$|H_A|=3^37^5=\sqrt{|\Lambda^{\#}/\Lambda|},$$

hence is maximal isotropic. Its inverse image is self-dual.

Exact power-basis reduction verifies $U_A=(\Sigma_A-I)/2\in M_2(\mathcal{O}_L)$; one has $U_A^2+U_A+I=0$ and $\mathcal{O}_K=\mathbb{Z}[U_A]$. On the $\mathfrak{p}_3$-part, $U_A\equiv 1$, while on the $\mathfrak{p}_7^5$-part it is a polynomial in Σ_A; hence H_A is $\mathcal{O}_K$-stable. The rational vector space, complex structure, K-action, and alternating form E_1 are unchanged by the lattice enlargement, so the stated Hodge-theoretic invariants are unchanged. □

Corollary 10.4 (The principal A-model loses the full vertex automorphism)**.** *No principally polarized self-dual overlattice*

$$\Lambda \subset \Lambda' \subset \Lambda^{\#}$$

*can be stable under the full normalized projective order-*7 *automorphism acting as* A_t^{T} *on homology.*

Proof. On the $\mathfrak{p}_3$-part the subgroup has order 27. Stability under A_t^{T} and its inverse implies stability under their sum $(t^3 - 3t)I = t^3 I$ in characteristic 3. Since $(t^3)^9 = t$ in $\mathbb{F}_{27}$, the subgroup is an $\mathbb{F}_{27}$-line. The characteristic polynomial of A_t is

$$X^2 - (t^3 - 3t)X + 1.$$

Its discriminant reduces to $1 - t$ modulo $\mathfrak{p}_3 = (3, t^3 - t^2 + t + 1)$, and exact finite-field arithmetic gives

$$(1 - t)^{13} = -1 \in \mathbb{F}_{27}.$$

Thus the discriminant is a nonsquare, the characteristic polynomial is irreducible, and A_t has no invariant residue-field line. □

11 Units, orientations and the maximal subpacket

The class-number and unit-norm arguments are also inputs to the integral geometry of both papers. Maximal-subpacket curve conclusions retain the packet execution premise; arithmetic orientation statements do not acquire an unrelated geometric-search assumption.

Theorem 11.1 (Collapse of the maximal-order integral conjugacy problem)**.** *Fix one of the* 16 *indices in Theorem 8.1, and let*

$$R_c = \mathcal{O}_L[\alpha_k] = \mathcal{O}_M, \qquad \alpha_k^2 - c_k\alpha_k + \pi = 0.$$

Then every matrix $A \in M_2(\mathcal{O}_L)$ *whose characteristic polynomial is* $X^2 - c_k X + \pi$ *and whose induced* $\mathcal{O}_L[\alpha_k]$*-lattice is locally free of rank one is* $\mathrm{GL}_2(\mathcal{O}_L)$*-conjugate to the companion matrix* A_{c_k} *in* (6.1)*. In particular, for these* 16 *traces the ordinary ideal-class part of the integral-conjugacy enumeration consists of a single class.*

The possible splitting of this single $\mathrm{GL}_2(\mathcal{O}_L)$*-class into* $\mathrm{SL}_2(\mathcal{O}_L)$*-classes is controlled by the finite quotient*

$$\mathcal{O}_L^\times / N_{M/L}(\mathcal{O}_M^\times). \tag{11.1}$$

More precisely, if $P \in \mathrm{GL}_2(\mathcal{O}_L)$ *conjugates* A *to* A_{c_k}*, then the determinant class of* P *modulo* $N_{M/L}(\mathcal{O}_M^\times)$ *is the only obstruction to replacing* P *by a determinant-one conjugator.*

Proof. Generalized Latimer–MacDuffee theory identifies integral similarity classes with isomorphism classes of rank-one lattices over the order R_c. Here $R_c = \mathcal{O}_M$. Masley's class-number-one conclusion for $\mathbb{Q}(\zeta_{21}) = M$ implies that every fractional $\mathcal{O}_M$-ideal is principal. Moreover $\mathcal{O}_M = \mathcal{O}_L[\alpha_k]$ is free of rank two over $\mathcal{O}_L$, with basis $(1, \alpha_k)$. Hence every admissible lattice is $\mathcal{O}_M$-isomorphic to $\mathcal{O}_M$, and multiplication by α_k is represented, after a change of $\mathcal{O}_L$-basis, by the companion matrix. This proves uniqueness up to $\mathrm{GL}_2(\mathcal{O}_L)$.

For the determinant-one refinement, the centralizer of multiplication by α_k is M, and its integral automorphisms are $\mathcal{O}_M^\times$. If a conjugator P is replaced by PC_ε, where C_ε represents multiplication by $\varepsilon \in \mathcal{O}_M^\times$, its determinant is multiplied by $\det_L(C_\varepsilon) = N_{M/L}(\varepsilon)$. Thus P can be adjusted to have determinant one exactly when $\det(P)^{-1}$ lies in the relative unit-norm image. Dirichlet's unit theorem gives equal rank 5 to $\mathcal{O}_L^\times$ and $\mathcal{O}_M^\times/\mu(M)$, so the quotient in (11.1) is finite. □

Lemma 11.2 (Class and narrow class numbers of L)**.** *The ordinary class number of L is $h(L) = 1$, the narrow class number is*

$$h^+(L) = 2,$$

and M/L is the narrow Hilbert class field. Moreover

$$[E_L : E_L^+] = 2^5, \qquad [E_L^+ : E_L^2] = 2.$$

Proof. The discriminant is $\operatorname{disc}(L) = 3^3 7^5 = 453789$. Minkowski's bound for a totally real sextic field is

$$\frac{6!}{6^6}\sqrt{\operatorname{disc}(L)} < 10.4.$$

Factoring the defining polynomial f modulo the rational primes below 11 gives

2	irreducible of degree 6
3	$(T^3 - T^2 + T + 1)^2$
5	$(T^3 - T - 2)(T^3 + T^2 + 2)$
7	$(T - 1)^6$.

Thus the only prime ideal of norm below 11 is the prime above 7, and it is principal because $N_{L/\mathbb{Q}}(t - 1) = f(1) = 7$. Hence $h(L) = 1$.

Let $\rho = \zeta_{21}$ and, for $a \in \{5, 11, 13, 17, 19\}$, set

$$\xi_a = \rho^{(1-a)/2}\frac{1 - \rho^a}{1 - \rho} = \frac{\sin(a\pi/21)}{\sin(\pi/21)} \in E_L.$$

These are real cyclotomic units. At the six real embeddings $k = 1, 5, 11, 13, 17, 19$, their signs, together with -1, are

	1	5	11	13	17	19
ξ_5	+	−	+	−	+	+
ξ_{11}	+	+	−	+	+	−
ξ_{13}	+	−	+	+	+	−
ξ_{17}	+	+	+	+	−	−
ξ_{19}	+	+	−	−	−	−
-1	−	−	−	−	−	−

and the corresponding vectors in $\mathbf{F}_2^6$ have rank 5. Hence the image of the unit-sign map has order at least 2^5, so $[E_L : E_L^+] \geq 2^5$.

The cyclotomic discriminant formula gives

$$|\operatorname{disc}(M)| = 3^6 7^{10} = \operatorname{disc}(L)^2,$$

so the relative discriminant of M/L is trivial. Thus the quadratic extension M/L is unramified at every finite prime and is contained in the narrow Hilbert class field of L; hence $h^+(L) \geq 2$. The standard narrow/ordinary class-number exact sequence gives

$$\frac{h^+(L)}{h(L)} = \frac{2^6}{[E_L : E_L^+]}.$$

Since $h(L) = 1$ and the denominator is at least 2^5, we have $h^+(L) \leq 2$. Therefore $h^+(L) = 2$, M is the narrow Hilbert class field, and $[E_L : E_L^+] = 2^5$. Finally $[E_L : E_L^2] = 2^6$ by Dirichlet's unit theorem, so $[E_L^+ : E_L^2] = 2$. □

Theorem 11.3 (Exact unit-norm image)**.** *Let* $E_M = \mathcal{O}_M^\times$ *and* $E_L = \mathcal{O}_L^\times$*. Then*

$$N_{M/L}(E_M) = E_L^+,$$

and hence

$$E_L/N_{M/L}(E_M) \cong E_L/E_L^+ \cong (\mathbf{Z}/2\mathbf{Z})^5.$$

Proof. The inclusion $N_{M/L}(E_M) \subseteq E_L^+$ is immediate from $\sigma(N_{M/L}(\varepsilon)) = |\widetilde{\sigma}(\varepsilon)|^2 > 0$ at every real place. For the reverse inclusion let $u \in E_L^+$. The quadratic extension M/L is cyclic and, by the discriminant calculation above, unramified at every finite prime. At a split finite place the local norm map is tautologically surjective, and at an unramified finite place the norm map on local units is surjective [5, Chap. III, Prop. 1.2, p. 98]. At each real place the norm $\mathbb{C}^\times \to \mathbb{R}^\times$ has image $\mathbb{R}_{>0}$, and u is totally positive. Thus u is a local norm everywhere. By the Hasse norm theorem for cyclic extensions, $u = N_{M/L}(x)$ for some $x \in M^\times$; see, for example, [7, Chap. VI, §4, Cor. (4.5), p. 384].

It remains to make the norming element a unit. Since u is a unit, the fractional ideal $I = (x)$ satisfies $I\bar{I} = \mathcal{O}_M$. Hence the valuations of I occur in opposite pairs at conjugate primes and vanish at conjugation-fixed primes. Choose a fractional ideal J with $I = J/\bar{J}$. Masley's class-number-one conclusion for $M = \mathbb{Q}(\zeta_{21})$ implies that $J = (y)$ for some $y \in M^\times$ [2, p. 179]. Then

$$\varepsilon = x\,\frac{\bar{y}}{y}$$

is a unit and $N_{M/L}(\varepsilon) = N_{M/L}(x) = u$. Therefore $E_L^+ \subseteq N_{M/L}(E_M)$, proving equality. The quotient statement follows from Lemma 11.2. □

Corollary 11.4 (No oriented splitting for holomorphic Hilbert-modular conjugacy)**.** *Fix one of the* 16 *maximal-order traces. If two corresponding matrices are* $\mathrm{GL}_2(\mathcal{O}_L)$*-conjugate by a matrix* P *with totally positive determinant, then they are already* $\mathrm{SL}_2(\mathcal{O}_L)$*-conjugate. Thus each maximal-order* $\mathrm{GL}_2^+(\mathcal{O}_L)$ *similarity class gives a single* $\mathrm{SL}_2(\mathcal{O}_L)$ *class relevant to the holomorphic action on* $\mathbb{H}^6$*.*

Proof. By Theorem 11.3, $\det(P)^{-1} = N_{M/L}(\varepsilon)$ for some $\varepsilon \in \mathcal{O}_M^\times$. Multiplication by ε lies in the integral centralizer and has L-determinant $N_{M/L}(\varepsilon)$. Replacing P by the correspondingly adjusted conjugator gives determinant one. A matrix acting holomorphically on every factor of $\mathbb{H}^6$ has totally positive determinant, so this is exactly the orientation class relevant to the Hilbert modular variety. □

Corollary 11.5 (Sharp arithmetic reduction for the strongest $d = 3$ candidates)**.** *For each of the* 12 *maximal-order, arithmetic-*$(3,3)$ *indices*

$$1, 11, 16, 17, 19, 20, 22, 32, 37, 38, 40, 41,$$

there is only one $\mathrm{GL}_2(\mathcal{O}_L)$ *integral similarity class to test. By Corollary 11.4, the holomorphically relevant* $\mathrm{GL}_2^+(\mathcal{O}_L)$ *class does not split further under* $\mathrm{SL}_2(\mathcal{O}_L)$*. Thus these* 12 *cases have no unresolved integral-conjugacy or unit-norm enumeration. The vertex computations below resolve eight of the twelve holomorphic orbit classes exactly; the remaining four are excluded by the full geometric Hilbert-modular orbit criterion* (6.5)*.*

Proposition 11.6 (Six maximal-order companion classes lie in normalized vertex orbits)**.** *Among the twelve maximal-order arithmetic-*$(3,3)$ *indices of Corollary 11.5,*

$$k \in \{11, 16, 19, 32, 37, 40\}$$

have companion representatives in normalized elliptic-vertex $\mathrm{SL}_2(\mathcal{O}_L)$*-orbits. More precisely,* $k = 11, 32$ *lie in the normalized* B*-vertex orbit and* $k = 16, 19, 37, 40$ *lie in the normalized* C*-vertex orbit. For* $k = 32, 37, 40$ *one uses the conjugate root* $c_k - \alpha_k$*, which has the same characteristic polynomial. Consequently only*

$$k \in \{1, 17, 20, 22, 38, 41\}$$

remain among the maximal-order arithmetic-$(3, 3)$ *packet for a separate geometric orbit test.*

Proof. Let T be the corresponding integral vertex endomorphism supplied by Proposition I.5.15, using B_t for $k = 11, 32$ and C_t for $k = 16, 19, 37, 40$, and write

$$A_{c_k} = \begin{pmatrix} 0 & -\pi \\ 1 & c_k \end{pmatrix}.$$

For any such T with characteristic polynomial $X^2 - c_k X + \pi$, set

$$P_0 = (e_1, Te_1).$$

Then $P_0 A_{c_k} = T P_0$. Exact rational-interval evaluation at the six real embeddings shows that $\det P_0$ is totally positive for

$$(B, 11),\ (B, \overline{32}),\ (C, 16),\ (C, 19),\ (C, \overline{37}),\ (C, \overline{40}),$$

where the bar means that T is replaced by $c_k I - T$. The determinant of P_0 need not be a unit (at the four displayed C-cases its absolute norm is 127), so P_0 is used only to determine the orientation class.

Indeed, any two L-conjugators between A_{c_k} and T differ on the right by an element of the centralizer $M^\times$. The ratio of their determinants is therefore $N_{M/L}(x)$ for some $x \in M^\times$, which is positive at every real place. Hence the six-sign vector of the determinant is independent of the chosen conjugator. For these six indices $\mathcal{O}_L[T] \cong \mathcal{O}_L[\alpha_k] = \mathcal{O}_M$ by Theorem 8.1; hence the induced $\mathcal{O}_M$-module on $\mathcal{O}_L^2$ is a torsion-free rank-one module over the Dedekind domain $\mathcal{O}_M$, and is therefore locally free. Theorem 11.1 consequently supplies an integral $Q \in \mathrm{GL}_2(\mathcal{O}_L)$ conjugating the same two matrices. The preceding sign calculation forces $\det Q$ to be a totally positive unit. Theorem 11.3 then writes $\det(Q)^{-1} = N_{M/L}(\varepsilon)$ for some $\varepsilon \in \mathcal{O}_M^\times$; multiplying Q by the corresponding integral centralizer element produces determinant one. Thus the companion matrix lies in the asserted normalized vertex $\mathrm{SL}_2(\mathcal{O}_L)$-orbit. □

Proposition 11.7 (The last imprimitive pair lies in the normalized A orbit)**.** *The canonical maximal-order companion classes for* $k = 20, 41$ *both lie in the normalized* A*-vertex* $\mathrm{SL}_2(\mathcal{O}_L)$*-orbit. The* $k = 41$ *realization uses the direct* A*-type action, while* $k = 20$ *uses its complex-conjugate CM action.*

Proof. Let $A_t = h_t A h_t^{-1}$ be the normalized A generator. Exact reduction in $\mathcal{O}_L = \mathbb{Z}[t]$ gives the integral matrix

$$Z_A = \begin{pmatrix} -t^5 + 7t^3 - 11t - 1 & -t^5 + 8t^3 - 15t - 2 \\ -t^4 + 6t^2 - 7 & t^5 - 7t^3 + 12t + 1 \end{pmatrix}$$

with

$$Z_A^2 - tZ_A + I = 0, \qquad Z_A^3 = A_t, \qquad Z_A^{21} = -I.$$

The element $4 - t^2$ is totally positive, and Z_A commutes with the elliptic matrix A_t; hence it fixes the same upper A-vertex point. Moreover $|N_{L/\mathbb{Q}}(t^2 - 4)| = 1$, so $\mathcal{O}_L[Z_A] = \mathcal{O}_M$.

Write

$$c_{20} = -t^3 + 2t - 2, \qquad c_{41} = -c_{20}.$$

For $\alpha_k = \zeta_{42}^k(1 + \zeta_{42} + \zeta_{42}^4)$, put

$$T_{20} = \alpha_{20}(Z_A^{-1}), \qquad T_{41} = \alpha_{41}(Z_A).$$

These integral matrices have characteristic polynomial $X^2 - c_k X + \pi$. If $A_{c_k} = \left(\begin{smallmatrix} 0 & -\pi \\ 1 & c_k \end{smallmatrix}\right)$, the explicit matrices

$$Q_{20} = \begin{pmatrix} 0 & 3t^5 - 17t^3 - 2t^2 + 23t + 3 \\ 1 & -2t^5 + 11t^3 + t^2 - 15t - 2 \end{pmatrix},$$

$$Q_{41} = \begin{pmatrix} 0 & 3t^5 - 17t^3 - 2t^2 + 23t + 3 \\ 1 & -2t^5 + 12t^3 + t^2 - 17t \end{pmatrix}$$

satisfy $Q_k A_{c_k} = T_k Q_k$. Their common determinant is the totally positive norm-one unit

$$q = -3t^5 + 17t^3 + 2t^2 - 23t - 3.$$

For a pair-specific determinant-one adjustment, take

$$\varepsilon = (1 - \zeta_{21}^{11})(1 - \zeta_{21}^{13}).$$

Exact cyclotomic multiplication gives $N_{M/L}(\varepsilon) = q^{-1}$ and, in the two companion bases, integral expressions $\varepsilon = x + y_k \alpha_k$. Thus $C_{\varepsilon,k} = xI + y_k A_{c_k}$ is integral, centralizes A_{c_k}, and has determinant q^{-1}. Consequently

$$G_k = Q_k C_{\varepsilon,k} \in \mathrm{SL}_2(\mathcal{O}_L), \qquad G_k A_{c_k} = T_k G_k.$$

The positive centralizer $C_{\varepsilon,k}$ fixes the unique upper companion point, while the positive intertwiner Q_k sends it to the unique upper fixed point of T_k, namely the normalized A vertex. Hence G_k gives the claimed holomorphic $\mathrm{SL}_2(\mathcal{O}_L)$ orbit equality.

Finally the full CM action, rather than the lower-left signs of the single Hecke matrix, determines the type. The assignments $\zeta_{42} \mapsto Z_A$ and $\zeta_{42} \mapsto Z_A^{-1}$ give respectively

$$\{1, 5, 17, 19, 29, 31\}, \qquad \{11, 13, 23, 25, 37, 41\},$$

which are the certified canonical types for $k = 41$ and $k = 20$. □

Theorem 11.8 (Hodge classification of the maximal-order $(3,3)$ packet)**.** *For the twelve maximal-order arithmetic-$(3,3)$ indices*

$$1, 11, 16, 17, 19, 20, 22, 32, 37, 38, 40, 41,$$

the canonical upper-half-plane companion CM types fall into exactly two $\mathrm{Gal}(M/\mathbb{Q})$*-orbits:*

$$\{1, 20, 22, 41\} \qquad \textit{and} \qquad \{11, 16, 17, 19, 32, 37, 38, 40\}.$$

The first orbit is imprimitive: its CM types have stabilizer $\{1, 29\}$ *and are induced from the sextic CM subfield* $\mathbb{Q}(\zeta_7)$*. Their rational Hodge algebras are generated by divisor classes. The second orbit is primitive; its Hodge eigenline counts in codimensions* $1, \ldots, 6$ *are*

$$6, 15, 22, 15, 6, 1,$$

and the only non-divisor eigenlines are the two lines of the $K = \mathbb{Q}(\sqrt{-3})$ Weil space.

For every one of the twelve companions, the K-Hermitian discriminant of the trace polarization is the class of -1. Consequently the Hodge conjecture holds for every one of the twelve associated polarized CM sixfolds: by Lefschetz $(1,1)$ in the imprimitive orbit and by Markman's theorem for the two Weil lines in the primitive orbit. Therefore the four companion classes not covered by Propositions 11.6 and 11.7 pose only a geometric curve-membership question, not a Hodge-algebraicity question.

Proof. For each index k, let Φ_k be the CM type selected by the upper fixed point of the companion matrix. Exact cyclotomic evaluation gives two Galois orbits of the displayed sizes. On the four-element orbit the stabilizer is $\{1, 29\}$; its fixed field is $\mathbb{Q}(\zeta_7)$, exactly as for the A-type in the earlier divisor-algebra calculation. Galois conjugacy therefore transfers the divisor-generated Hodge algebra to all four types. On the eight-element orbit the stabilizer is trivial. The representative $\{1, 5, 23, 25, 29, 31\}$ equals $31\Phi_C$ modulo 42, so Lemma I.5.10 and its translation invariance give $6, 15, 22, 15, 6, 1$ Hodge eigenlines and exactly two non-divisor middle-degree lines, the two K-Weil eigenspaces. The two registered exact enumerations verify this conceptual classification for the full primitive packet. The imprimitive four-element orbit still uses its separate exact degree-two and product enumeration.

Finally write $\sqrt{-3} = x_k + y_k\alpha_k$ over L. Exact reduction in $\mathbb{Q}[t]/(f)$ gives

$$N_{L/\mathbb{Q}}(y_k) = -1728 = -2^6 3^3$$

for every one of the twelve indices. The Mobius representative is $S_k = x_k I + y_k A_{c_k}$ and the actual action is S_k^{T}. In the K-basis $(0, t^j)$ the Gram matrix is $(-\operatorname{Tr}_{L/\mathbb{Q}}(y_k t^{i+j}))_{0\le i,j\le 5}$. Its rank is six, so the determinant of the associated Hermitian form is

$$\operatorname{disc}(L) N_{L/\mathbb{Q}}(y_k) = -2^6 3^6 7^5.$$

The factors $2^6 3^6 7^5$ are norms from K up to the remaining negative sign, so the discriminant class is -1. The claimed algebraicity follows from the preceding Hodge-algebra descriptions and Markman's theorem in the primitive case. □

Proposition 11.9 (Exact exclusion of the four nonvertex maximal-order companions)**.** *The canonical companion quotient points for*

$$k \in \{1, 17, 22, 38\}$$

do not lie on the normalized McMullen curve.

Proof. McMullen's doubled-triangle construction gives a compact convex labeled quadrilateral in each of the six normalized coordinates. Exact vertex invariants and rational Sturm intervals give strict outward lower height and upper modulus bounds. If a companion point is carried into all six quadrilaterals by an element of $\mathrm{SL}_2(\mathcal{O}_L)$, the entries of that matrix lie in explicit archimedean boxes.

The positive trace form $\operatorname{Tr}_{L/\mathbb{Q}}(x^2) = \sum_i \sigma_i(x)^2$, together with an exact rational LDL^{T} decomposition, enumerates every integral point in the resulting finite ellipsoids. Determinant one is imposed exactly, with a separate complete $c = 0$ branch. Final geodesic-side predicates are elements of L and are signed by rational root intervals, retaining exact boundary zeros.

For each of $k = 17, 38$, the search produces 2,307 coupled bottom rows and 9,812 final matrices; for each of $k = 1, 22$, it produces 1,849 bottom rows and 9,072 final matrices. No matrix survives all six coordinate quadrilaterals. Separate SymPy/rational and standard-library/Sturm implementations reproduce the complete matrix/failure records and canonical digests for all four indices.

This establishes avoidance of the original product only. The determinant-one companion transfers in the proof of Theorem 9.10 identify the two labelled points as masks $2, 13$. That theorem also supplies their complete reflected-product exclusions. Lemma I.3.5 then gives the claimed full-curve conclusion; the original-product calculation alone is insufficient. □

Corollary 11.10 (Complete maximal-order geometric packet)**.** *Among the twelve maximal-order arithmetic-*$(3,3)$ *companion classes, eight lie in normalized elliptic-vertex orbits—*$k = 20, 41$ *at* A*,* $k = 11, 32$ *at* B*, and* $k = 16, 19, 37, 40$ *at* C*—and the remaining four are excluded by Proposition 11.9. No geometric orbit-membership case remains within these twelve selected canonical companions. The stronger all-orientation statement is Theorem 9.10.*

12 Comparison with the proposed Hecke reduction

The comparison is subsidiary to the positive results. The following summary is proved by the detailed arguments in this section and the polarization computation in Paper I. It does not identify the constructed principal family with a different moduli normalization.

Theorem 12.1. *For the Hecke construction of [6, Sec. 5], the following corrections hold.*

(i) A degree-ℓ *Hecke isogeny on an abelian sixfold cannot satisfy* $\alpha^\dagger\alpha = [\ell]$*; in the rank-two Hilbert modular model one has instead* $\alpha^\dagger\alpha = (\det_L \alpha)I = \pi I$ *and characteristic polynomial* $X^2 - cX + \pi$*.*

(ii) For a prime $\mathfrak{P}$ *of* M *above* $\mathfrak{l} = (\pi)$*, the relative norm is* $N_{M/L}(\mathfrak{P}) = \mathfrak{l}$*, while its absolute norm is* 43*; these are not interchangeable.*

(iii) The published count $44 \cdot 64 = 2816$ *does not by itself eliminate the unrestricted variable* $\delta \in \mathrm{SL}_2(\mathcal{O}_L)$ *appearing in a Hecke lift.*

(iv) The six Galois components of McMullen's modular embedding have distinct Schwarz exponent data and are not obtained from one common hypergeometric equation merely by changing monodromy.

(v) For this field the quotient $\mathbb{H}^6/\mathrm{SL}_2(\mathcal{O}_L)$ *with its standard* $\mathcal{O}_L^2$ *lattice carries the explicit polarization of type* $(1, 7, 7, 21, 21, 21)$*, not a principal one. A principal Hilbert–Blumenthal moduli interpretation requires a different polarization module/arithmetic group; Proposition* I.4.3 *constructs a principally polarized model only after an isogeny/lattice change.*

(vi) The "CM isolation" and "origin of the Hodge–Weil classes" obstructions asserted in [6, Obs. 4.6] are not hypotheses of Markman's algebraicity theorem. Markman's Theorem 1.5.1 applies to every polarized abelian sixfold of Weil type with discriminant -1*, including CM fibers. Thus, once the polarized Weil structure and discriminant* -1 *are verified, neither CM endomorphisms nor the absence of a separately exhibited secant presentation obstruct application of the theorem.*

Items (i)–(iv) concern the explicit Hecke-search mechanism. Item (v) reaches the moduli-theoretic setup used earlier in [6]: the quotient by $\mathrm{SL}_2(\mathcal{O}_L)$ *with the standard* $\mathcal{O}_L^2$ *lattice is not itself the asserted principally polarized family. The lattice modification in Theorem* I.6.4 *gives a principally polarized family after passage to a connected finite cover and an isogeny. It changes the integral moduli data. The present paper therefore does not treat Mostaed's earlier finiteness argument as literally unaffected by the normalization. Item (vi) concerns only the stated obstruction to using*

Markman's theorem; it does not supply the missing discriminant for an arbitrary point of the intersection.

The results in this part are logically secondary to Theorem I.1.2. They explain why the local determinant π and the corrected cyclotomic norm equation are the appropriate arithmetic inputs, and they delimit exactly which claims of [6] are affected. The earlier finiteness theorem for an intersection point already lying in the relevant locus is not challenged here.

12.1 Applicability of Markman's sixfold theorem

[6, Observation 4.6] states that Markman's theorem does not apply to a CM fiber in the Weil locus for three independent reasons: CM isolation, the origin of the Weil classes, and uncontrolled discriminant. Only the last of these is an actual hypothesis issue for Theorem 1.5.1.

First, a CM point is special, but it is not deformation-rigid as a polarized abelian variety of Weil type: it is a point on the positive-dimensional Weil moduli component determined by $(n, K, \det H)$, and nearby deformations simply cease to have the extra CM endomorphisms. The fact that Markman's proof propagates algebraicity through a deformation family therefore does not impose a non-CM hypothesis on the theorem's final statement.

Second, the Hodge–Weil space is intrinsic to the K-action:

$$W_K(A) = \bigwedge_K^6 H^1(A, \mathbb{Q}) \subset H^6(A, \mathbb{Q}).$$

A secant-sheaf construction is a method used in the proof of algebraicity; it is not additional structure required in the statement of the theorem. Markman's Theorem 1.5.1 states that the Hodge–Weil classes of *polarized abelian sixfolds of Weil type* with discriminant -1 are algebraic, with no exclusion of CM points and no hypothesis that the given fiber itself be presented by a chosen secant sheaf.

Consequently, for an arbitrary point of the proposed intersection the uncontrolled discriminant remains a genuine obstacle to invoking Markman. For the explicit vertex constructed in Part I, however, Proposition I.5.8 computes the discriminant to be -1, so Theorem 1.5.1 applies directly. This distinction is important: the proof of Theorem I.1.2 does not need to deform the explicit CM point while preserving its full CM endomorphism algebra.

12.2 The degree obstruction in Proposition 5.2

We first isolate a contradiction that does not depend on any subtle theory of CM types.

Lemma 12.2. *Let A be an abelian variety of dimension g and let $[n] : A \to A$ denote multiplication by n. Then*

$$\deg[n] = n^{2g}.$$

Proof. This is the standard degree formula for multiplication on an abelian variety; see Milne [4, Thm. 7.2]. □

Proposition 12.3 (Degree obstruction)**.** *Let A be an abelian sixfold and $\phi : A \to B$ an isogeny of degree $\ell > 1$. If $\phi^\dagger$ denotes the Rosati adjoint with respect to principal polarizations on A and B, then*

$$\phi^\dagger \phi \neq [\ell].$$

In particular, the identity used in the proof of [6, Prop. 5.2] cannot hold in dimension six.

Proof. Duality preserves the degree of an isogeny, and conjugating the dual by principal polarizations does not change degree. Hence

$$\deg(\phi^\dagger\phi) = \deg(\phi^\dagger)\deg(\phi) = \ell^2.$$

If $\phi^\dagger\phi = [\ell]$, Lemma 12.2 with $g = 6$ would give

$$\ell^2 = \deg(\phi^\dagger\phi) = \deg[\ell] = \ell^{12},$$

impossible for $\ell > 1$. □

Remark 12.4. The familiar formula "dual isogeny composed with the isogeny equals multiplication by the degree" is an elliptic-curve formula. In higher dimension, an isogeny of degree d does admit a complementary isogeny whose composition is $[d]$, but that complementary map has degree d^{2g-1} in general; it is not the polarization adjoint of degree d.

The contradiction matters because Proposition 5.2 immediately uses the asserted identity to derive a quadratic equation $X^2 - cX + \ell = 0$. That constant term is incompatible both with Proposition 12.3 and with the determinant-π matrix model used later in the same section.

12.2.1 The exact polarization-adjoint identity in the Hilbert modular model

There is a direct replacement for the erroneous identity. Let $V = L^2$ and equip it with the standard alternating L-bilinear form

$$\langle x, y\rangle_L = x^{\mathsf{T}} J y, \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The rational Riemann form of a Hilbert–Blumenthal abelian variety is obtained from such an L-valued form by composing with a trace $\mathrm{Tr}_{L/\mathbb{Q}}(\xi\,\cdot)$ for a suitable totally positive polarization parameter ξ; because every matrix considered below is L-linear, the scalar ξ does not change the adjoint formula.

Lemma 12.5 (Adjoint equals adjugate)**.** *For*

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(L),$$

the adjoint with respect to $\langle\cdot,\cdot\rangle_L$ *is*

$$A^\dagger = J^{-1}A^{\mathsf{T}}J = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \mathrm{adj}(A).$$

Consequently

$$A^\dagger A = (\det_L A) I_2. \tag{12.1}$$

Proof. This is a direct multiplication. The defining identity $\langle Ax, y\rangle_L = \langle x, A^\dagger y\rangle_L$ gives $A^\dagger = J^{-1}A^{\mathsf{T}}J$, and for a 2×2 matrix this is its classical adjugate. Multiplying the adjugate by A gives $(\det A)I_2$. □

Corollary 12.6 (Correct Hecke similitude factor)**.** *In the standard Hilbert modular realization, if a Hecke lift α has $\det_L(\alpha) = \pi$, then on rational homology, and hence in* $\mathrm{End}^0(A)$*,*

$$\boxed{\alpha^\dagger \alpha = \pi}.$$

Moreover $\deg(\pi) = |N_{L/\mathbb{Q}}(\pi)|^2 = \ell^2$*, exactly matching* $\deg(\alpha^\dagger\alpha) = \ell^2$*.*

Proof. Equation (12.1) gives the first assertion. Since $H_1(A, \mathbb{Q})$ has rank two as an L-vector space, multiplication by $r \in L$ has rational determinant $N_{L/\mathbb{Q}}(r)^2$, which is its isogeny degree for integral totally positive r. Taking $r = \pi$ and $N(\pi) = \ell$ gives ℓ^2. □

Corollary 12.6 is stronger than a degree objection: it identifies the precise element that replaces the rational scalar ℓ. It also agrees identically with the determinant convention written in Section 5.3 of [6].

12.3 Repair from the Hilbert-modular determinant

The statement that Hecke fixed points are CM can be repaired without the erroneous dual-isogeny identity. Corollary 12.6 supplies the correct Rosati relation, while Cayley–Hamilton supplies the same constant term independently.

Proposition 12.7 (Correct local characteristic polynomial)**.** *Let $\mathfrak{l} = (\pi)$ be a prime of $\mathcal{O}_L$ with a totally positive generator π. Suppose a Hecke fixed point admits a lift represented on the rank-two L-module $H_1(A, \mathbb{Q})$ by*

$$\alpha = \delta\gamma_j \in M_2(L), \qquad \det_L(\alpha) = \pi.$$

Put $c = \mathrm{Tr}_L(\alpha)$, the 2×2 matrix trace. Then

$$\alpha^2 - c\alpha + \pi = 0. \tag{12.2}$$

If α is non-scalar, $L[\alpha]$ is a quadratic L-algebra with norm π.

Proof. Cayley–Hamilton on the two-dimensional L-vector space gives

$$X^2 - (\mathrm{tr}_L\,\alpha)X + \det_L(\alpha).$$

The determinant is π by construction, yielding (12.2). If α is non-scalar, its minimal polynomial has degree two and coincides with this polynomial. □

Proposition 12.8 (CM repair)**.** *Assume in addition that, for all six real embeddings $\sigma_i : L \hookrightarrow \mathbb{R}$, the matrix $\sigma_i(\alpha)$ fixes a point of $\mathbb{H}$. Then the discriminant*

$$\Delta = c^2 - 4\pi$$

is totally negative. Consequently $L(\alpha)$ is a totally imaginary quadratic extension of the totally real field L, hence a CM field of degree 12 over $\mathbb{Q}$.

Proof. A real 2×2 matrix of positive determinant that fixes a point of the upper half-plane is elliptic (or scalar). In the non-scalar case its characteristic polynomial has nonreal conjugate roots, so its discriminant is negative. Applying this at every real embedding gives

$$\sigma_i(c)^2 - 4\sigma_i(\pi) < 0$$

for every i. Thus Δ is totally negative and adjoining a root of $X^2 - cX + \pi$ gives a totally imaginary quadratic extension of L. □

Remark 12.9. A scalar lift cannot have determinant ideal equal to a prime ideal with odd valuation: if $\alpha = rI_2$, then $(\det \alpha) = (r)^2$. Hence the prime-determinant situation excludes the scalar case.

Thus the conclusion of Mostaed's Proposition 5.2 is repairable, but the proof and the arithmetic invariant must be changed. In particular, if the repaired quadratic field is the prescribed

$$M = L(\sqrt{-d}),$$

and $\alpha = a + b\sqrt{-d}$, then

$$N_{M/L}(\alpha) = a^2 + db^2 = \pi, \tag{12.3}$$

not the diagonal rational integer ℓ.

12.4 Relative norm, absolute norm, and integrality

There is a second independent issue in Section 5.6 of [6]. Let $\mathfrak{P}$ be a prime of $\mathcal{O}_M$ lying over $\mathfrak{l} \subset \mathcal{O}_L$. If $\mathfrak{l}$ splits in M/L, then

$$N_{M/L}(\mathfrak{P}) = \mathfrak{l},$$

whereas the absolute ideal norm is

$$N_{M/\mathbb{Q}}(\mathfrak{P}) = \ell.$$

These have different codomains. If $\mathfrak{P} = (\varpi)$ and $\mathfrak{l} = (\pi)$, then

$$(\varpi\bar{\varpi}) = \mathfrak{l} = (\pi),$$

so

$$\varpi\bar{\varpi} = u\pi \tag{12.4}$$

for a unit $u \in \mathcal{O}_L^\times$. A normalization can absorb suitable norm units, but the result is an element of L generating $\mathfrak{l}$, not automatically the rational integer ℓ.

There is also an integrality subtlety. Even when $\alpha \in \mathcal{O}_M$ and $\alpha = a + b\sqrt{-d}$ with $a, b \in L$, it need not follow that $a, b \in \mathcal{O}_L$. The order $\mathcal{O}_L[\sqrt{-d}]$ need not equal $\mathcal{O}_M$. Therefore conditions such as

$$c^2 - 4\pi \in -4d\,(\mathcal{O}_L)^2$$

are generally stronger than the field-theoretic condition

$$c^2 - 4\pi \in -d\,(L^\times)^2.$$

The integral structure should be imposed in $\mathcal{O}_M$ itself rather than by assuming a power-basis decomposition with integral coefficients.

12.5 A separate gap in the claimed 2816-system reduction

The determinant correction does not address another logical problem in the claimed finite enumeration.

Section 5.3 writes a fixed-point lift as

$$\alpha = \delta\gamma_j, \qquad \delta \in \mathrm{SL}_2(\mathcal{O}_L), \tag{12.5}$$

with j running through $\ell + 1 = 44$ Hecke cosets. The component fixed-point quadratics depend on the four entries of α, and hence on the unrestricted matrix δ. Immediately afterwards, Step 5 counts only the 44 cosets and the 2^6 choices of roots of the six quadratics, producing $44 \cdot 64 = 2816$ systems [6, Sec. 5.3].

Proposition 12.10 (The stated count does not follow from the stated parametrization)**.** *Equations* (12.5) *and the* 44 *Hecke cosets do not by themselves produce* 44 *concrete fixed-point systems. The coefficients of each component quadratic still vary with the infinite set* $\mathrm{SL}_2(\mathcal{O}_L)$*. Therefore an additional elimination argument, a modular-polynomial equation, or an independent effective bound on* δ *is required before a finite count such as* 2816 *is justified.*

Proof. For fixed j, the matrix γ_j is fixed but δ is not. The coefficients a, b, c, d in the fixed-point equation are the entries of $\delta\gamma_j$, so changing δ changes the equation itself. Choosing one of the two roots of each of six already-specified quadratics is a finite operation, but the quadratics are not specified until δ has been fixed or eliminated. No such elimination or finite enumeration of δ is supplied in the stated Step 5. Thus the product $44 \cdot 2^6$ counts root choices conditional on a matrix, not all matrices that may realize a fixed point. □

This proposition should be read narrowly: it says that the *published derivation* of the number 2816 is incomplete. It does not prove that no alternative finite algorithm exists. Indeed, Hecke correspondences are algebraic correspondences of finite degree, so one expects a finite algebraic formulation after passing to appropriate modular equations; that formulation must still be supplied.

12.6 Correction of the hypergeometric stage

The preprint's Section 5.4 contains an additional, logically independent problem. McMullen's modular embedding satisfies

$$f_i(g_1 z) = g_i f_i(z), \tag{12.6}$$

where g_j denotes the image of g under the j-th real embedding [3, Sec. 6, Eq. (6.1)]. By contrast, equation (4) of [6, Sec. 5.4] places g_i on both the source and target. The later displayed functional equation for the generator B in the same section uses g_1 on the source and is therefore consistent with McMullen rather than with the stated equation (4).

More importantly, the assertion that all six component maps are governed by the same hypergeometric equation with parameters

$$(a, b; c) = \left(\frac{19}{42}, \frac{3}{7}; \frac{13}{14}\right)$$

cannot describe the six Galois-conjugate projective monodromy representations. The reason is already visible from local monodromy. The parent triangle group has projective local exponent differences

$$\lambda = \frac{1}{14}, \qquad \mu = \frac{1}{21}, \qquad \nu = \frac{1}{42}.$$

Under the real embedding indexed by $k \in \{1, 5, 11, 13, 17, 19\}$, the relevant roots of unity are sent to their k-th powers. Hence the projective exponent differences become

$$\lambda_k = \left\| \frac{k}{14} \right\|, \qquad \mu_k = \left\| \frac{k}{21} \right\|, \qquad \nu_k = \left\| \frac{k}{42} \right\|, \tag{12.7}$$

where $\|x\|$ denotes distance to the nearest integer. The Schwarz parameters satisfying

$$1 - c_k = \lambda_k, \qquad c_k - a_k - b_k = \mu_k, \qquad a_k - b_k = \nu_k$$

are therefore

$$c_k = 1 - \lambda_k, \qquad a_k = \frac{1 - \lambda_k - \mu_k + \nu_k}{2}, \qquad b_k = \frac{1 - \lambda_k - \mu_k - \nu_k}{2}. \tag{12.8}$$

Proposition 12.11 (Six distinct Schwarz equations)**.** *The six Schwarz parameter triples, in the standard embedding order, are*

k	$(a_k, b_k; c_k)$
1	$(19/42,\ 3/7;\ 13/14)$
5	$(11/42,\ 1/7;\ 9/14)$
11	$(2/7,\ 1/42;\ 11/14)$
13	$(3/7,\ 5/42;\ 13/14)$
17	$(1/2,\ 2/21;\ 11/14)$
19	$(1/2,\ 1/21;\ 9/14)$.

In particular, only the identity embedding has the parameter triple displayed in Section 5.4 of [6].

Proof. The eigenvalue ratio of an elliptic element of projective order n is an n-th root of unity. A cyclotomic embedding indexed by k raises this ratio to its k-th power. Projectively, the corresponding exponent difference is defined modulo 1 and up to sign, so a priori each local difference may be represented by either r or $1-r$. The representative in $(0, 1/2]$ used in (12.7) is forced here by hyperbolicity, not merely chosen by convention.

Indeed, for each $k \in \{1, 5, 11, 13, 17, 19\}$, enumerate the 2^3 sign representatives at the vertices of orders $14, 21, 42$. A hyperbolic Schwarz triangle must have total normalized angle strictly less than 1. Exactly one of the eight triples has sum < 1, namely

$$\left(\left\| \frac{k}{14} \right\|, \left\| \frac{k}{21} \right\|, \left\| \frac{k}{42} \right\| \right).$$

Moreover, in every one of the six cases the smallest rejected alternative has sum exactly 1, hence is Euclidean. Concretely, the surviving sums are

$$\frac{1}{7}, \frac{5}{7}, \frac{20}{21}, \frac{16}{21}, \frac{17}{21}, \frac{19}{21}$$

for $k = 1, 5, 11, 13, 17, 19$, respectively, while the nearest alternative in each case has sum 1. Thus the hyperbolic Galois triangle uniquely selects the nearest-integer representatives in (12.7). Substitution into (12.8) gives the stated table. □

There is also a conceptual obstruction to the "one common equation" formulation. For a fixed second-order linear ODE, the ratio attached to any two choices of basis of its two-dimensional solution space differs by a constant Möbius transformation. Thus if all six Schwarz maps were ratios of bases of one and the same hypergeometric equation on the same base coordinate, their projective monodromy representations would be conjugate in $\mathrm{PGL}_2(\mathbb{C})$. McMullen's construction of the curve uses a subgroup whose diagonal Galois image is Zariski dense in $\mathrm{SL}_2(\mathbb{R})^6$ [3, Sec. 5 and proof of Thm. 1.9]; such pairwise conjugacy would force the image into a proper graph subgroup. The six distinct triples in Proposition 12.11 are the local-monodromy manifestation of this non-Shimura phenomenon.

Accordingly, the appropriate analytic construction is, up to the vertex-normalizing Möbius transformations,

$$f_i = S_i \circ S_1^{-1}, \tag{12.9}$$

where S_i is the Schwarz map for the i-th parameter triple in Proposition 12.11. This turns the distinguished-lift test (6.4) into a numerically implementable problem: reduce $w_1(c)$ to the first fundamental triangle, evaluate its hypergeometric coordinate via S_1^{-1}, and then evaluate the five distinct Schwarz maps S_i with the corresponding target-vertex normalizations.

12.7 Dependency audit of the preprint

Table 1 records what the present corrections do and do not affect.

Table 1: Dependency audit of the relevant claims in [6].

Item	Status after audit	Reason
Sections 1–4; finiteness of $V \cap \mathcal{W}_K$	Requires a specified principal-moduli normalization	The Section 5 determinant correction alone does not refute the abstract finiteness argument. Its application to the named principally polarized locus still requires the integral moduli comparison discussed in Remark I.1.4.
Proposition 5.2, conclusion "Hecke fixed point is CM"	Repairable	The published proof uses an impossible degree identity, but the standard polarization adjoint gives $\alpha^\dagger\alpha = \pi$, and the determinant-$\pi$ matrix model plus ellipticity gives a replacement proof (Corollary 12.6 and Propositions 12.7–12.8).
Proposition 5.2, equation $X^2 - cX + \ell$	Incorrect for the stated prime-ideal Hecke lift	The same section later has $\det(\delta\gamma_j) = \pi$; Cayley–Hamilton gives constant term π.
Section 5.3 generator π_1	Unsuitable if total positivity is required	π_1 has two negative real embeddings. Equation (3.5) supplies a totally positive generator of the same prime.
Section 5.3 fixed-point equations	Structurally usable after normalization	The Möbius fixed-point equation itself is standard, but its determinant and lift must be normalized consistently.
Theorem J / 2816 systems	Not established as written	The infinite variable $\delta \in \mathrm{SL}_2(\mathcal{O}_L)$ remains in the coefficients; $44 \cdot 64$ does not eliminate it.
Section 5.6 norm equation $a^2 + db^2 = \ell$	Incorrect for a fixed prime ideal $\mathfrak{l}$	The corrected relative norm is a generator π of $\mathfrak{l}$, up to the appropriate norm-unit normalization.
Claim $N_{M/L}(\varpi) = N(\mathfrak{P}) = \ell$	Incorrect identification of norms	Relative ideal norm is $\mathfrak{l}$; absolute ideal norm is ℓ.
Requirement $a, b \in \mathcal{O}_L$ for every integral $\alpha \in \mathcal{O}_M$	Not generally valid	$\mathcal{O}_M$ need not equal $\mathcal{O}_L[\sqrt{-d}]$; integrality must be checked in $\mathcal{O}_M$.

Item	Status after audit	Reason
Examples 5.7–5.8	Correct identities for the diagonal norm 43, but not the corrected local Hecke norm equation	$4+3\sqrt{-3}$ and $6+\sqrt{-7}$ have relative norm 43, whereas a fixed $\mathfrak{l}$-Hecke lift has norm π with $N_{L/\mathbb{Q}}(\pi) = 43$.
Section 5.4 hypergeometric stage	Incorrect as stated	The source/target equivariance is misstated, and the six Galois components have distinct local exponent data; Proposition 12.11 gives the corrected parameter triples.
Passage from Y_L to X_L	Published shortcut is incomplete; explicit repair available	McMullen first constructs the curve in $(\pm\mathbb{H})^6/\mathrm{SL}_2(\mathcal{O}_L)$. The replacement $f_i \mapsto -\overline{f_i}$ does not preserve the original target representation in general. Proposition 7.1 computes the actual sign vector and shows that $h_t = \mathrm{diag}(t,1) \in \mathrm{GL}_2(\mathcal{O}_L)$ transfers the curve integrally to X_L.
Geometric nonemptiness for $K = \mathbb{Q}(\sqrt{-3})$	Resolved for the normalized trace-polarized fibers; literal principal-locus comparison open	Theorem I.5.3 and Proposition I.5.7 identify the normalized B-vertex and its actual $(3,3)$ CM type. The isogenous principal models do not establish membership in Mostaed's named principal image.

The most important consequence of the audit is therefore not that the earlier finiteness theorem is false. Rather, the explicit Hecke *search mechanism* in Section 5 needs to be rebuilt around the local determinant π. Proposition 6.1, generalized Latimer–MacDuffee theory, and Theorem 8.1 provide a concrete route to replacing the unrestricted lift variable by finitely many integral conjugacy classes.

13 Computer-assisted results and reproducibility

The arguments use computer assistance only for finite exact calculations. No theorem is based on an uncertified floating-point equality. The following inventory describes the packet certificates. The additional fixed-field Hodge-extension evidence is specified in Section H.2.6 and Sections H.3–H.4; these form a separate regression scope and do not enlarge the historical 1405-command execution claim below. A clean immutable-commit replay remains a separate release gate.

Source	Role

certificates/packet_order_coverage_v58.py; independent/packet_order_coverage_stdlib_v58.py	Two exact paths for the 84 norm-fiber representatives, 42 traces, nine embedded orders and affine-unit identities in Proposition 8.3. Completeness of the norm fiber and the common maximal population also use the human proofs.
certificates/known_cm_comparison_v58.py	Exact regression for the supplied geometric CM types and explicit actions in Proposition I.5.18; this does not independently reconstruct the geometric types.
certificates/k11_vertex_bridge_v60.py; independent/k11_vertex_bridge_stdlib_v60.py	New exact cyclotomic and quadratic reconstructions of the integral determinant-one $k = 11$ bridge in Theorem I.5.3, including strict ellipticity. These replace the missing historical computations.
certificates/d3_completeness_v60.py; independent/d3_completeness_quadratic_v60.py; certificates/d3_matrix_manifest_v60.jsonl	Separate exact reconstructions of the 1280 trace-labelled matrix representatives in Theorem 8.6. The human module and unit proof establishes completeness; these files give no ambient-point count.
certificates/d3_points_v60.py; independent/d3_points_quadratic_v60.py; certificates/d3_points_v60/canonical_points_v59.json; certificates/d3_points_v60/matrix_to_point_bridges_v59.jsonl	Two exact paths for the 144 canonical points, unit-square basis corrections and all 1280 integral determinant-one transports in Proposition I.3.2. The two-root and intrinsic multiplier-order proof establishes quotient distinctness.
certificates/full_packet_arithmetic_v62.py; independent/full_packet_cyclotomic_v62.py; certificates/full_packet_v62/ambient_points.json; certificates/full_packet_v62/root_bridges.jsonl	Separate quadratic and degree-12 cyclotomic reconstructions of the complete 3520-class, 368-point census in Theorem I.3.3, including the finite local quotients, all module and orientation representatives, and 736 root bridges. No curve-membership test is included.
certificates/a_vertex_certificate_v31.py; independent/a_vertex_hodge_node_v31.js	Exact A-type, discriminant, and Hodge calculation; the Node path is a supplied-type combinatorial cross-check.
certificates/a_principal_local_v33.py; certificates/three_vertex_level_v37.py	Restricted local free-line checks; the homology correction and nonfree common subgroup are certified below.
certificates/principal_homology_v56.py; independent/principal_homology_stdlib_v56.py	Two separate exact homology reconstructions: integral K-actions, corrected Hermitian bases, principal special fibers, and the common nonfree subgroup of Proposition I.6.1. The global family is treated separately below.
certificates/principal_descent_v57.py; independent/principal_descent_stdlib_v57.py	Exact stabilizer indices 224 and 28, selected CM-order index 27, and surviving vertex powers. Theorem I.6.4 supplies the human descent and algebraization argument.
certificates/vertex_saturation_v35.py	Exact B/C saturation, C-type, discriminant, and Hodge calculation.

certificates/maximal_vertex_orbits_v38.py; certificates/four_vertex_audit_v40.py	Exact maximal B/C orientations and four-vertex audit.
certificates/packet_hodge_v39.py; independent/packet_cm_hodge_sturm_v41.py	Two exact independent packet CM-type/Hodge implementations; the second is the theorem-grade standard-library/Sturm verifier required for Theorem 11.8.
independent/packet_hodge_node_v40.js	Floating-point regression evidence only; it is not a theorem certificate or an independent exact implementation.
independent/quadrilateral_k17_stdlib_v41.py	Standard-library quotient-field, Sturm, interval, and normalized-quadrilateral foundation used by the independent exact paths.
certificates/residual_orbit_k17_k38_v42.py; independent/residual_orbit_k17_k38_sturm_v42.py	Two exact independent exhaustive exclusions for $k = 17, 38$.
certificates/residual_orbit_k1_k22_v43.py; independent/residual_orbit_k1_k22_sturm_v43.py	Two exact independent exhaustive exclusions for $k = 1, 22$.
certificates/a_vertex_orbits_k20_k41_v44.py; independent/a_vertex_orbits_k20_k41_sturm_v44.py	Two exact independent A-orbit implementations, including the direct/conjugate CM-action assignment.
certificates/d7_nonmaximal_classes_v29.py; certificates/d7_ambient_points_v45.py; independent/d7_ambient_points_sturm_v45.py	Exact 512-class and 112-point ambient classification. The number 107 is the v45 checkpoint, 75 is the v53 checkpoint, and Corollary 9.12 closes that $d = 7$ remainder.
certificates/d7_r83_mask14_v46.py; independent/d7_r83_mask14_sturm_v46.py	Retained first proper-R_{83} exact exclusion pair; the registered v47 run subsumes this primary search.
certificates/d7_r83_masks14_30_v47.py; independent/d7_r83_masks14_30_sturm_v47.py	Two exact independent paths for masks $14, 30$.
certificates/d7_r83_masks15_31_v48.py; independent/d7_r83_masks15_31_sturm_v48.py	Two exact independent paths for masks $15, 31$.
certificates/d7_r83_masks6_22_v49.py; independent/d7_r83_masks6_22_sturm_v49.py	Two exact independent paths for masks $6, 22$.
certificates/d7_r83_masks1_17_v50.py; independent/d7_r83_masks1_17_sturm_v50.py	Two exact independent paths for masks $1, 17$.
certificates/d7_r83_batch10_v51.py; independent/d7_r83_batch10_sturm_v51.py	Two exact independent paths for ten reflected pairs/twenty points.
certificates/d7_r83_masks13_29_v52.py; independent/d7_r83_masks13_29_sturm_v52.py	Two exact independent paths for masks $13, 29$.

certificates/d7_r83_masks7_23_v53.py; independent/d7_r83_masks7_23_sturm_v53.py	Two exact independent paths for masks $7, 23$, including the complete $1955 = 1954 + 1$ bottom-row split.
certificates/d7_remaining_maximal_v61.py; independent/d7_remaining_maximal_sturm_v61.py; certificates/d7_remaining_maximal_expected_v61.py	Two exact paths for all eleven remaining maximal masks: 89,544 candidates per path, including 1,284 with $c = 0$, and zero survivors. These computations concern the original product; the next entry supplies the required second target for Theorem 9.10.
certificates/reflected_primary_v83.py; independent/reflected_sturm_v83.py; certificates/reflected_packet_v83/expected.json; certificates/reflected_packet_v83/selection.json	Two separate full-only reflected-target paths for the thirteen nonhit maximal labels: 1,113,924 candidates per path, including 12,188 with $c = 0$, and zero survivors. The complete mathematical dependency closure, source relocation and exact frozen plans are included; the expected table binds eight canonical artifacts and all nine result fields for every label.
certificates/d7_r41_v63.py; independent/d7_r41_sturm_v63.py; certificates/d7_r41_expected_v63.py	Two exact exhaustive paths for all 32 proper-$(41, t - 12)$ points: 62,474 bottom rows and 265,212 candidates per path, including 33,468 with $c = 0$, and zero survivors. The frozen table binds every center's counts and digests. Theorem 9.11 and Corollary 9.12 give the scoped exclusion and complete $d = 7$ classification.
certificates/d7_r83_principal_v55.py; independent/d7_r83_principal_sturm_v55.py; certificates/d7_r83_principal_expected_v55.py	Two exact independent exhaustive paths for all 32 principal-Picard-class proper-R_{83} points; the third source is the frozen comparison table.
certificates/broader_principal_v75.py; independent/broader_principal_sturm_v75.py; certificates/broader_principal_v75/expected.json	Two exact exhaustive paths for the six named principal-Picard mask-zero points in Proposition 9.14: 54,120 candidates per path and zero survivors. The portable data directory contains the point and unit witnesses; the expected table binds every point's counts and eight canonical artifact hashes. No other mask or whole-stratum exclusion is asserted.
certificates/r127_pic1_v80.py; independent/r127_pic1_sturm_v80.py; certificates/r127_pic1_v80/expected.json	Fresh exhaustive reconstruction of the nonprincipal `R127:pic1:00` row of Proposition 9.14: 8,488 candidates and zero survivors in each path. The entry-unit witness file is in the same directory; the point and DEN/NUM witnesses are reused from the preceding portable package. The principal `R127:pic0:00` replay is an already-certified control, not another new exclusion.
certificates/continuous_primary_v82.py; independent/continuous_sturm_v82.py	Original-target portable replay: 249 nonvertex labels in both paths, with six full historical controls, for 504 commands.

certificates/reflected_primary_v83.py; independent/reflected_sturm_v83.py; certificates/reflected_packet_v83/selection.json; certificates/reflected_packet_v83/expected.json; certificates/reflected_packet_v83/source-relocation.json	Reflected-target 365-label exclusion certificate, with complete plans and both exact paths. The common 371-record table also includes the six labelled positive controls (A, B, C in both targets), producing $730 + 12$ full commands.

The two broader exclusion packages recompute the finite sets from their portable source/data closure and read no discovery or build-directory search output at runtime. Their full success status requires all counts and all eight canonical artifacts to agree with the frozen comparison tables. Source-only and preparation-only checks make no geometric claim. The targeted full replays and their recorded adversarial audits are complete; an immutable whole-suite release remains a separate gate.

Principal-model regressions. The following paired implementations verify the exact inputs to the principal-model arguments. Their human completeness arguments are given in the cited propositions. They are additional regression tests and are not part of the historical 1405-command packet execution record. Each pair has a corresponding test in the repository's test directory; the installed regression replays both implementations.

Source	Role
certificates/principal_unmarked_bc_v122.py; independent/principal_unmarked_bc_v122.py	Actual-lattice reconstruction and the complete finite norm obstruction for unmarked polarized distinction.
certificates/principal_bc_isogeny_v123.py; independent/principal_bc_isogeny_v123.py	Complete scalar-multiplier-2 map list, common kernel, and inputs to the full polarized automorphism-group argument (Proposition I.7.5).
certificates/principal_bc_unpolarized_v125.py; independent/principal_bc_unpolarized_v125.py	Full multiplier order, maximal-order saturations and local residue obstruction to unpolarized isomorphism (Proposition I.7.6).
certificates/principal_bc_min_degree_v126.py; independent/principal_bc_min_degree_v126.py	Degree-7 cyclic-kernel witness and small-prime norm exclusions (Proposition I.7.7).
certificates/principal_bc_hom_v127.py; independent/principal_bc_hom_v127.py	Complete Hom module, fractional-ideal products and source pullback identities (Proposition I.7.8).
certificates/principal_order_class_v129.py; independent/principal_order_class_v129.py	Residue field, explicit unit norms and unit-image inputs for the ordinary Picard-group computation (Proposition I.7.9).

Whole-packet execution provenance and regression scope. The original-target cover is $116 + 249 = 365$ distinct nonvertex points. Its legacy component is assembled from 150 retained original commands, 63 individually accepted single-mask replacements and two complete later replacements. The 65 replacement commands partition nine old command domains. This is a reviewed equivalent cover, not a claim that the interrupted old 159-command pool or the old 65-job

recovery pool completed normally. Failed and interrupted histories remain retained. The original-target portable aggregate is bound by `experiments/continuous_v82_review/v82_outcome_review.json`.

Four historical V42/V43 companion commands have no recorded per-command end-source guard; their use assumes the archived sources and runtime were not transiently replaced during execution. The original legacy pool also lacks a common final source/runtime guard. Present hashes cannot establish the absence of a transient replacement, and no missing guard is reconstructed. The previously scoped maximal-only acceptance remains recorded in `experiments/continuous_v82_review/om13_q26_scoped_outcome_review.json`.

The reflected aggregate selects exactly 730 full backend results: 557 original results and 173 selected continuation results. It reconciles all 365 paired empty outcomes, complete row journals and additional completed duplicates, while retaining failed predecessors. Its actual outcome and retention are bound by `experiments/continuous_v82_review/v93_actual_conditional730_outcome_retention_review.json`. One continuation cohort halted after 78 complete results, with 26 interrupted and 69 unstarted tasks. Its selected 26 results and 52 completed duplicates retain their individual driver checks; the missing coordinator-wide final guards are not supplied by later hashes. Use of this cohort retains the explicitly reviewed ordinary historical source/runtime immutability assumption.

The portable mathematical closure has sixteen files: fourteen unchanged files and two drivers with reversible root/helper-path relocation. The nineteen-file executed closure includes its operational wrappers. Whole-executable comparison covers dynamic imports, active target installation, plans, globals and actual interpreter/package profiles. The old expected-data module imported by a retained foundation remains an inert lineage dependency for these searches: its historical assertions may abort execution, but it supplies no new-label bounds or candidate decisions. New comparison expectations are checked only after complete enumeration.

The portable controls comprise the original sixteen negative labels plus the exact additional `R127:pic1:03` pair, hence 34 full negative runs, and twelve strict positive runs on the final common 371-record package. The added pair exercises the otherwise uncovered identity-denominator scaling branch. The original 32-run conjunction failed that observed-interface requirement and remains a failed record; it is not renamed as a pass. The revised conjunction retains every original result comparison and requires coverage of the unchanged observed interface inventory. Controls test relocation and entrypoint integration; they do not replace exhaustive results for the other labels.

The final adoption joining the two target populations, the selected executions, and the source correspondence is `experiments/continuous_v83/whole_packet_assessment/final-adoption.json`. This is the final packet receipt; the intermediate outcome receipts above are supporting records, not substitutes for that join. Its use retains the packet execution hypothesis stated in Paper I.

The installed registration and evidence map are recorded in `tests/registration/combined1405.json` and `tests/registration/combined1405-installation.json`. The inventory is exactly $159+504+12+730 = 1405$ commands. Within the reflected 730, the 34 control commands precede the remaining 696; the old maximal-only 26 commands are subsumed rather than duplicated. The existing 159 command vectors and default four-worker limit are preserved. Installation, equivalent-cover adoption and targeted executions are not a fresh integrated 1405-command suite pass. An immutable source/evidence release, its final manuscript compilation and visual PDF inspection remain separately recorded verification steps.

The audited local environment uses Python 3.14.6, SymPy 1.14.0, and Node.js 22.19.0. The file `requirements.txt` pins the SymPy version. The Sturm implementations written with the Python standard library do not import SymPy. The verifier `scripts/run_certificates.py` fails

if any registered source is missing, if a registered program exits nonzero, or if the required Node runtime is unavailable. The wrapper `verify_all.sh` also checks repository invariants, compiles the manuscript when a LaTeX engine is available, writes the full run log, and emits a SHA-256 manifest. Separate implementations are machine-verification paths, not independent human proofs.

All theorem-relevant embedding labels in the SymPy certificates are ordered by exact rational endpoints of pairwise disjoint root intervals, with strict separation asserted. Decimal formatting in the independent quadrilateral module is display-only. The exact packet verifiers exhaust the CM-character subsets as a regression check; Lemma I.5.10 supplies the conceptual classification for the primitive block.

Availability of computational evidence. The manuscript and research repository are currently private working materials. The historical receipts and source paths in this supplement refer to that retained workspace. A complete, versioned package sufficient to inspect every theorem-relevant dependency has not been released. The selected source snapshot and its short smoke checks are not such a package. No assertion of public data availability or fresh full-suite reproduction is made. Before external circulation, the evidence required by each computer-assisted result must be assembled and its access instructions checked against the exact manuscript version.

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